

Representation, Innovation, and Product Adoption*

Jennifer Kao
UCLA Anderson

Tamar Oostrom
The Ohio State University and NBER

August 2, 2025

Abstract

During research and development, firms must decide which user groups to engage when developing and testing their products. We investigate how subsidizing representation shapes innovation, product quality, and adoption. In 2000, Medicare expanded coverage of clinical trial costs, reducing financial constraints for elderly enrollees, a historically underrepresented group. Reducing financial constraints leads to an increase in representation: trial sponsors broaden eligibility criteria and enroll a higher share of elderly participants. We find an immediate and large increase in the number of clinical trials for diseases common among the elderly and increased drug utilization among elderly individuals. To explore potential mechanisms, we analyze clinical trial results, adverse events, and drug approvals. Our findings show how financial incentives can increase representation and lead to substantial changes in the rate and impact of subsequent innovation.

JEL Classifications: O31, O38, I14, I18, J14

Keywords: Research and Development, Experimentation, Inequality, Regulation, Elderly

*We are grateful to seminar participants at the Berkeley Health Economics Workshop, the Becker Friedman Health Economics Initiative Annual Conference, Electronic Health Economics Colloquium, HBS Junior Innovation Economics Conference, Junior(ish) Health Economics Summit, NBER Productivity, Innovation, and Entrepreneurship Program Meeting, Toulouse School of Economics, UCLA, University of Kansas, University of Toronto, USC, and Washington State University for helpful feedback on this project. We thank clinical research finance experts at UCLA for helpful discussions. We thank Grace Nussle for excellent research assistance. All errors are our own.

1 Introduction

During research and development (R&D), firms often conduct tests to generate information on product quality and market demand (Kerr, Nanda and Rhodes-Kropf, 2014; Camuffo et al., 2020; Koning, Hasan and Chatterji, 2022). An important decision is which users to engage in product development and testing (Cao, Koning and Nanda, 2023). For instance, software developers must choose platform users to evaluate product performance. Similarly, pharmaceutical firms must decide which patient populations to include in clinical trials to assess drug safety and efficacy, an important decision due to the high costs of recruitment and enrollment (Sertkaya et al., 2016). These decisions shape the type of information firms generate, the products they bring to market, and how those products are ultimately used.

For firms, ensuring representation (defined here as ensuring that characteristics of test users mirror those of the target market) may support the development of products that effectively target these users and encourage adoption (Alsan et al., 2024). However, underrepresentation of specific demographic groups persists. For example, elderly individuals (i.e., individuals aged 65 and older) make up nearly 63 percent of new cancer cases but only comprise 25 percent of participants in cancer clinical trials (Hutchins et al., 1999). Financial constraints often limit firms’ ability to recruit representative users (Lara et al., 2001; Herrera et al., 2010).

In this paper, we examine how financial constraints shape representation in R&D and, importantly, how shifts in representation affect innovation, product adoption, and quality. We study a 2000 executive memorandum requiring Medicare to cover the routine patient costs associated with clinical trial participation. The memorandum aimed to reduce financial barriers that limited the enrollment of older adults, a large but historically underrepresented group in product development (Herrera et al., 2010).

We first estimate the memorandum’s impact on elderly representation in clinical trials. We then examine how shifts in elderly representation subsequently impact the rate of clinical trials, prescription utilization, and drug quality. To do so, we exploit cross-disease variation in exposure to the memorandum using a difference-in-differences (DID) design, leveraging the fact that diseases more prevalent among the elderly were more affected by the memorandum. We assemble a novel

dataset of clinical trials, with newly collected information on participant enrollment criteria and actual participant demographics, which we link to detailed information on drug utilization, clinical trial outcomes, and adverse events. We present four key findings.

First, we show that reducing financial constraints significantly shifts representation in product development. Following the memorandum, firms increase the number of trials whose enrollment criteria included elderly participants. There was no significant change in trials that allowed non-elderly participants. We also find suggestive evidence that the memorandum increases the share of actual elderly enrollees. Our analysis spans 1995 to 2005, when Medicare did not cover prescription drug costs outside of those administered in a physician’s office or hospital. Medicare Part D was signed in December 2003 and implemented in January 2006. Importantly, we find that the change in representation occurs immediately after the memorandum, before Medicare Part D was signed into law or implemented.¹

Second, greater representation is accompanied by a significant increase in innovative activity. Diseases most prevalent in the elderly experienced a relative increase of 24 percent in the annual number of clinical trials and a 29 percent increase in the annual likelihood of a trial, compared to less prevalent diseases. This pattern is consistent with the view that reducing financial constraints made it easier to include elderly participants, increasing firms’ incentives to invest in trials targeting those populations. We find that an immediate increase in clinical trials is largely driven by drugs that had previously undergone clinical testing. These effects are larger in diseases with a high cost of clinical trials and larger in the United States (where the memorandum was enacted) than in Europe.

Third, increased representation also leads to increased product adoption. Clinical trial information, which can be disseminated through public registries, publications, advertisements, and drug labels, can potentially change the product adoption decisions of consumers or physicians (McKibbin, 2023). We find that, following the memorandum, there is a disproportionate increase in drug utilization among elderly participants for diseases common among the elderly, relative to other diseases and non-elderly participants. Exploiting variation across diseases in exposure to the memorandum

¹The passage of prescription drug coverage for Medicare had been debated for over two decades, and the passage of the final bill was a surprise with thin margins (Alpert, 2016).

and patient age, we show that similar shifts in utilization occur among individuals likely to already have drug insurance who would be less affected by the anticipation of Medicare Part D.

Importantly, these increases in utilization are driven by drugs tested in clinical trials after the memorandum, relative to a placebo group of drugs that were not clinically tested during our sample period. At the disease level, we find that diseases with larger increases in trials after the memorandum also had larger increases in prescriptions, further highlighting the link between clinical trial activity and drug adoption.

Finally, to better understand the mechanisms driving increases in drug adoption, we examine whether greater representation leads to improvements in product quality, which encompasses the safety and effectiveness of a drug for its intended user group. Using the precision of trial research outcomes and post-market adverse events, we find suggestive evidence that increased representation improves both the perceived and actual quality of drugs, though these results are not statistically significant.

Taken together, the results show how reducing financial barriers to increase demographic representation has significant consequences for the rate and impact of subsequent innovation. These findings are economically meaningful. For example, we observe an average of 1.09 Phase II and III trials within a disease each year. [Sertkaya et al. \(2016\)](#) estimate that the average cost of a clinical trial is \$4.35 million, suggesting that the increase due to the Medicare Memorandum of 2000 spurred clinical investments of more than \$1 million annually per disease common among the elderly. This estimate is conservative, not accounting for the additional profits from faster participant enrollment and increased drug utilization.

Additionally, we do not find evidence that these clinical investments increased regulatory approvals for novel drugs; most drugs that were newly tested in clinical trials were already approved. Instead, additional trials seem to be conducted to support new indications, test the drug against competitors, or learn more about secondary outcomes and risks. We observe suggestive evidence that these trials facilitated learning through failures, reflected in increased suspensions, withdrawals, and discontinuations. This fits with evidence that the memorandum increased innovation in areas with weaker private financial incentives ([Eisenberg, 2005](#)).

Our paper contributes to the literature on innovation, diversity, product testing, and the role of

financial barriers in shaping R&D outcomes.

While prior work has largely focused on the demographics of individuals leading R&D efforts (Nielsen et al., 2017; Koning, Samila and Ferguson, 2021; Einiö, Feng and Jaravel, 2019), we shed light on the consequences of representation among test users (i.e., the individuals from whom information about product quality is generated). Most related is Cao, Koning and Nanda (2023), which examines how gender-concordance between early users and the target market on an online platform is associated with more product upvotes and pageviews. Similarly, Alsan et al. (2024) provides survey evidence that race-based representation in clinical trials influences physicians’ reported prescribing intentions for Black Americans. We contribute real-world, causal evidence from a setting where trial enrollment decisions are a costly component of pharmaceutical firms’ R&D investments (Sertkaya et al., 2016). Linking data on trial participants and post-approval prescriptions, we show that increased elderly representation among test users leads to greater clinical trial activity and product adoption. Our findings highlight representation as a meaningful lever through which firms shape both the rate and impact of subsequent innovation.

Second, we contribute to research on the drivers of underrepresentation in innovation (Gompers and Wang, 2017). A growing literature has documented that trust (Alsan and Wanamaker, 2017), stereotypes (Hebert, 2025), and bias (Kanze et al., 2018; Ewens and Townsend, 2020) contribute to gender and racial disparities in innovative outcomes. Several recent papers show that researcher characteristics can shape the composition of test users (Gupta, 2022; Alsan et al., 2025). Related work by Michelman and Msall (2022) examines the removal of FDA guidance on women of childbearing age and finds limited effects on female-focused clinical trials, suggesting that regulatory guidance alone may be insufficient to shift representation. We highlight the importance of financial constraints in shifting trial participation among underrepresented groups. This builds on evidence that access to financing shapes who innovates (Hegde and Tumlinson, 2014; Guzman and Kacperczyk, 2019; Howell and Nanda, 2024).

Third, the literature on pharmaceutical innovation has largely focused on how changes in expected revenue affects innovation (see, e.g., Blume-Kohout and Sood, 2013; Agha, Kim and Li, 2022). These papers have found that changes in market size changes the willingness of firms to invest in innovation. This paper is the first to examine the other half of this investment decision:

whether reductions in the costs or burdens of conducting clinical trials influence firms' innovation decisions. We also complement prior work documenting the utilization impacts of Medicare Part D (e.g. [Duggan and Morton \(2010\)](#); [Zhang et al. \(2009\)](#); [Lichtenberg and Sun \(2007\)](#)) by showing that increases in elderly prescription use began years earlier, following a separate policy that reduced participation barriers for clinical trials.

Finally, we focus on the elderly, an important but underrepresented demographic in both product development and academic research, which has largely focused on gender and racial disparities (see, e.g., [Einiö, Feng and Jaravel, 2019](#); [Truffa and Wong, 2025](#); [Luo and Zhang, 2025](#)). While non-financial constraints (e.g., trust, bias) may be key participation barriers for some underrepresented groups, financial and logistical barriers appear to be particularly salient for the elderly ([Lara et al., 2001](#); [Herrera et al., 2010](#)). For firms and policymakers, these barriers may be more directly addressable: firms can reduce participation costs by sharing recruitment platforms or investing in flexible trial infrastructure (e.g., systems that support remote data collection), and policymakers can offer targeted subsidies to spur recruitment. This study shows that addressing these financial barriers may spur representation in the R&D process, and ultimately increase innovation and product adoption.

2 Institutional Background

2.1 Drug Development

In the US, drug development involves clinical trials and costs rise at each phase. Drugs that successfully demonstrate safety in Phase I trials proceed to Phase II trials in which their efficacy is tested in a few hundred participants. If successful, the drugs move to Phase III trials in which their efficacy is tested in several thousand participants. Upon successfully completing multiple Phase III trials, the sponsor can submit a New Drug Application to the FDA for final approval. After approval, drugs can be used for other indications ("off-label") without additional regulatory review.

2.2 Costs of Clinical Research

The drug development process is costly (typically costing a drug manufacturer \$800 million) and long (often taking between 8 and 12 years) (DiMasi, Grabowski and Hansen, 2016; DiMasi, Hansen and Grabowski, 2003). Drug manufacturers aim to enroll eligible participants quickly, due to competition in participant recruitment and to minimize the high fixed costs associated with extended trial recruitment periods. An average Phase III trial with direct costs of \$11 million is estimated to have \$310,000 in patient recruitment costs, \$2,250,000 for clinical procedures for patients, and \$850,000 for lab costs for patients (Sertkaya et al., 2016). The Medicare memorandum indirectly reduced recruitment costs via its awareness program and fully covered the cost of routine care, including most clinical procedures and labs.

Before 2000, the costs of clinical procedures and labs were often shared between drug manufacturers and participants (Pear, 2000). While there is limited information on the actual allocation of costs and substantial variation, drug manufacturers typically covered patient recruitment, routine clinical trial expenses, and data analysis (Aaron and Gelbrand, 2000). Routine costs account for a large share of the cost of most clinical trials and include clinical procedures for patients, diagnostic and laboratory tests, post-surgical care, room and board for hospitalized patients, and office visits (Aaron and Gelbrand, 2000). Participants often received treatments at minimal cost but faced expenses for some medical care, screening, and travel. Medicare occasionally covered some trial costs but frequently deemed trials experimental and denied coverage. Before 2000, Medicare did not cover routine patient costs. In 2000, about 17 percent of Medicare patients had a Medicare Advantage plan, and private insurers also did not historically cover these costs.

The uncertainty surrounding the financial responsibility associated with clinical trials deterred participation and led to unexpected costs for enrollees (Pear, 2000). Clinical trial investigators could not assure elderly prospective participants that Medicare would cover trial costs, a risk that was required to be disclosed during the informed consent process (Aaron and Gelbrand, 2000). Additionally, Medicare’s policy of not routinely preauthorizing care and its legal status as a payer of last resort meant that trial participants incurred significant costs if Medicare denied the manufacturer’s claims. Trial participants had to be informed of these financial risks as part of the informed consent

process. Taken together, this created substantial uncertainty for participants and manufacturers. To address some of this uncertainty, about one-third of all industry trials were fully funded and covered all routine costs without billing insurance.²

2.3 Representation of the Elderly In Clinical Trials

The high costs and uncertainty of insurance coverage may have contributed to the limited participation of elderly participants in clinical trials. Although individuals aged over 65 constitute 16.8 percent of the U.S. population, they account for just 6 percent of participants in the average pre-2000 trial in our dataset (U.S. Census Bureau, 2023). Individuals aged over 65 have been reported to be underrepresented in diseases such as cancer, diabetes, heart failure, and osteoporosis (Hutchins et al., 1999; Lau et al., 2022).

Beyond cost concerns, the elderly encounter logistics, comorbidities, health literacy, and cognitive barriers to trial participation (Herrera et al., 2010). For drug manufacturers, a homogeneous trial population reduces the variance of treatment effects, lowers the sample size needed for statistical power, and thus makes the inclusion of the elderly potentially more costly.

2.4 Medicare Memorandum of 2000

In June of 2000, President Clinton ordered Medicare to start covering most of the costs of clinical trials for its beneficiaries (largely patients aged 65 and older). The memorandum stated, “Medicare will immediately begin to reimburse for the routine patient care costs, and costs due to medical complications associated with participation in a clinical trial.” This memorandum included items or services typically provided absent a clinical trial, such as conventional care during a trial. This also includes monitoring the effects of the investigational drug or service, items required for the provision of the investigational drug or service, and costs related to the diagnosis and treatment of complications. Drug manufacturers would continue to pay for the collection and analysis of data and the investigational item or service itself.³

²Source: Author’s interviews with clinical trial managers.

³See National Coverage Determination (NCD) for Routine Costs in Clinical Trials (310.1)

Vice President Al Gore, speaking about the memorandum, said, “Speeding up enrollment can accelerate the discovery and use of cost-saving, life-saving, new therapies” (Pear, 2000). This memorandum also included an informational campaign to inform Medicare beneficiaries, doctors, hospitals, and other health care providers of the new policy. The intended impact of this policy was to decrease uncertainty and increase subsidization of clinical trial costs for both elderly participants and drug manufacturers. Specifically, patients were assured that their routine costs were covered and that manufacturers could bill Medicare instead of risking covering costs themselves.⁴

3 Data

We are interested in measuring how the Medicare Memorandum of 2000 shifted trial enrollment, innovation, drug adoption, and quality for a given disease. We define diseases using the International Classification of Diseases, version 9 (ICD-9).⁵ Our final dataset consists of 390 ICD-9 codes.

1. **High Medicare Share (“Old”) Diseases:** Following a commonly used approach (see, e.g., Acemoglu et al., 2006; Duggan and Morton, 2010; Blume-Kohout and Sood, 2013), we proxy for exposure to the memorandum by calculating the share of individuals covered by Medicare associated with each ICD-9. We use the 1996 (the earliest year in which data is available) to 1999 MEPS, a nationally representative survey of the U.S. civilian non-institutionalized population (Medical Expenditure Panel Survey, 1996-2010). We categorize diseases as having a “high Medicare share” if they have more than the median share of Medicare individuals with that diagnosis from 1996-1999. We show that our results are similar when using a continuous measure of Medicare share. Appendix Figure A.1 presents the distribution of Medicare shares across diseases. For example, low Medicare share diseases (“young diseases”) include attention-deficit/hyperactivity disorder (3 percent Medicare share). In contrast, high Medicare share diseases (“old diseases”) include Parkinson’s disease (81 percent Medicare share).

2. **Clinical Trials:** To create a dataset of clinical trials and participant enrollment, we begin

⁴While the program primarily aimed to increase elderly participation in trials, it also sought to ensure that the trials generated information necessary for determining Medicare reimbursement for the drugs being tested.

⁵Examples of diseases at the ICD-9 code level include “malignant neoplasm of the colon” and “diabetes mellitus.”

with data from the Cortellis Clinical Trials Intelligence Database (“Cortellis”) (Cortellis Clinical Trials Intelligence Database, 1995-2005). Cortellis contains clinical trial data from clinical trial registry websites, press releases, financial filings, and FDA submissions. This dataset contains each trial’s enrollment criteria, start year, duration, phase, results, and associated ICD-9 codes. We restrict the sample to trials that begin between 1995 and 2005 since data are sparse in earlier years. We restrict trials to those with at least one U.S. location, as the memorandum was U.S.-based.⁶ Finally, we restrict our sample to Phase II and III trials as Phase I trials are not well-reported and are arguably not covered by the memorandum.⁷ As private-sector and public-sector funders may have different objective functions, we restrict our analysis to trials with any private-sector funding. This results in a final dataset of 4,964 trial-diseases.

3. **Inclusion Criteria and Enrollment:** We incorporate ClinicalTrials.gov data on clinical trial enrollment characteristics, such as the number of elderly (aged 65 and over) and non-elderly participants (see Appendix Figure A.2 for an example). We supplement this data with enrollment counts from publications for a random subset of trials with a PubMed identifier. We also analyze the text of the inclusion criteria for each trial. Each clinical trial has specific enrollment criteria listing eligible participants (e.g. “Female”, “Ages 18-44”, “Type 2 diabetes”). Enrollees are required to meet these criteria. We update our enrollment counts based on the enrollment criteria when available. For example, a trial with enrollment criteria restricted to ages 18-64 should have no participants over age 65.
4. **Drug Utilization:** We collect data on drug utilization from the 1996 to 2010 MEPS surveys. We categorize drug utilization for individuals aged 65-75 and separately for those aged 50-60. We link MEPS drug names to clinical trials in the Cortellis data, which provides information on the diseases tested, the number of clinical trials for that drug over time, and the drug’s approval status by indication. We assign each drug to its initial FDA-approved disease; we

⁶We drop trials that occur in other countries, since we do not take a stand on whether international pharmaceutical investment is a complement or substitute for U.S.-based investment. We show a placebo test using European trials in Section 5.1.1.

⁷Covered trials must evaluate efficacy, not just safety. See National Coverage Determination (NCD) for Routine Costs in Clinical Trials (310.1)

use the most common disease for that drug in Cortellis if there are several initial approvals. As above, we identify high and low Medicare share diseases based on the share of Medicare patients with that disease diagnosis. Consequently, a prescription drug in MEPS can be categorized as treating a disease with either a high or low Medicare share. See Appendix Section A.1 for additional details.

5. **Disease Characteristics:** We characterize diseases in two ways. To determine the under and over-representation of elderly individuals across diseases, we compute each ICD-9’s average disease burden by incorporating data from The Global Burden of Disease (GBD) dataset (Global Burden of Disease, 2019). Disease burden is measured with noise, aggregated from a variety of sources and frequently unavailable, so we use a 2019 cross-section instead of historical data.⁸

We also estimate clinical trial costs. Because detailed ICD-9 code-level estimates of clinical trial costs are not publicly available, we use two proxies: (1) the average trial duration and (2) whether a disease is primarily associated with outpatient or inpatient care. Longer trials are likely to incur higher costs, and inpatient care is generally more costly and invasive compared to outpatient care (Cox et al., 2011; Friedrichs et al., 2016). We divide diseases by whether they are above or below the median in trial duration or prevalence in MEPS inpatient files.

6. **Drug Quality:** We proxy for drug quality in two ways. First, we measure the precision of clinical trial results using the number of trials with statistically significant outcomes on ClinicalTrials.gov. Second, we measure adverse events included in the Adverse Event Reporting System (AERS), a database of reported adverse events used to monitor post-marketing safety by the FDA. We obtained historical AERS files from 1969-2006 through a Freedom of Information Act request initiated by the Investigative Reporters and Editors group (FDA Adverse Events Reporting System, 1969-2006). Adverse events can be reported by health care professionals and consumers such as patients, family members, lawyers, and others. AERS reports the drug name, event year, age and gender of the individual who experienced the event.

⁸Disease burden is an imperfect proxy of a disease’s potential market size, since market size is disease burden interacted with access to health care and willingness to pay for treatment.

7. Drug Development Status The Cortellis data contain information on the last known development status for each drug over time. We define a drug as being approved if it is registered or launched in the U.S.. We define a drug as having stopped development if it is suspended, discontinued, or withdrawn. We construct measures of drug development that include the number of approvals or terminations in each disease quarter.

We construct a panel dataset at the disease-year level. Table 1 shows that from 1995 to 2005, a higher number of clinical trials’ enrollment criteria included elderly participants: the average disease-year had 0.93 trials that included elderly individuals, compared to only 0.15 trials limited to non-elderly participants. In terms of actual enrollment, elderly individuals make up only 13 percent of enrolled participants per trial. Overall, each disease-year averaged 1.09 clinical trials. To investigate drug utilization, we also construct a dataset at the patient-year level. On average, patients aged 65-75 have 1.05 prescriptions for young diseases and 2.85 unique prescriptions for old (high Medicare share) diseases.

4 Impact on Representation

4.1 Disparities in Clinical Research

Before examining the causal impact of the Medicare Memorandum, we begin with descriptive evidence of changes in elderly representation. In particular, we describe how elderly representation in clinical trial enrollment compares to elderly disease burden and how this relationship changes over time.

For each ICD-9 code, we calculate both the share of elderly participants in clinical trials and the share of the disease burden that affects the elderly. Figure 1 compares trial enrollment and disease burden at the ICD-9 code-level. The 45-degree line represents perfect parity between disease burden and trial enrollment. From 1995-1999, all diseases fell below the 45-degree line, which corresponds to elderly underrepresentation in trials. However, this pattern changes after 2000: the slope between elderly disease burden and trial enrollment increases from 0.065 to 0.467, which represents greater age-based parity. The difference in slope between 1995-1999 and 2001-2005 is statistically significant ($p < 0.001$), showing that age-based disparities in trial enrollment were

reduced after the memorandum.

4.2 Empirical Strategy

We measure the causal effect of the Medicare Memorandum of 2000 on representation in clinical trials using a DID design. Our baseline specification compares outcomes in ICD-9 codes with an above median share of Medicare recipients (“high Medicare share”) versus a below the median share, before and after the memorandum:

$$y_{jt} = \sum_t \beta_t \mathbf{1}\{quarter = t\} \times \mathbf{1}\{MedicareShare_j\} + \mathbf{1}\{MedicareShare_j\} + \gamma_{jt}\delta_{c(j)} + \delta_j + \delta_t + \epsilon_{jt} \quad (1)$$

where y_{jt} denotes the trial-based outcome (e.g., the number of trials enrolling elderly participants) for ICD-9 disease code j in calendar quarter t . The variable $\mathbf{1}\{MedicareShare_j\}$ is an indicator for whether disease j had an above-median share of Medicare patients prior to 2000. The coefficients β_t capture changes in the outcome for high Medicare share diseases relative to low Medicare share diseases over time. The regression controls for $\mathbf{1}\{MedicareShare_j\}$, disease fixed effects (δ_j), and quarter fixed effects (δ_t). To allow for time-varying trends across broader disease categories, we include interactions of linear quarter with the ICD-9 disease chapter, which is a broader ICD-9 categorization ($\delta_{c(j)}$) than ICD-9 codes.⁹ Standard errors are clustered at the ICD-9 code level.

In our event studies, we set the reference period as the two years prior to the memorandum and constrain the *average* of these coefficients to be zero. Relative to omitting a single period prior to treatment, using a wider reference period yields a uniform level shift of the coefficients. This approach helps improve precision, as a single reference point may be noisily estimated, but does not change the relative time patterns or the interpretation of any pre-trends (Miller, 2023). Appendix Figure A.3 shows that patterns are similar using the standard choice to omit just the period prior to treatment.

⁹For example, while an ICD-9 code might represent a specific condition such as “malignant neoplasm of the colon,” the corresponding disease chapter would capture a broader category like “neoplasms.”

In regression specifications, we modify equation (1) to include an indicator for years after the Medicare Memorandum ("Post") interacted with $\mathbf{1}\{MedicareShare_j\}$. We estimate equation (2) at the year-level to simplify interpretation:

$$y_{jt} = \mathbf{1}\{Post\} \times \mathbf{1}\{MedicareShare_j\} + \mathbf{1}\{MedicareShare_j\} + \gamma_{jt}\delta_{c(j)} + \delta_j + \delta_t + \epsilon_{jt} \quad (2)$$

In addition to a binary measure of *MedicareShare*, we present our main estimates using a continuous measure of Medicare share.

4.3 Results on Representation

4.3.1 Trial Enrollment Criteria

To understand how the Medicare Memorandum shapes representation in product development, we begin by examining whether it changes drug manufacturers' enrollment criteria. Each clinical trial has specific inclusion and exclusion criteria, which cover age, diagnoses, health, and prior treatment. We categorize a trial as prioritizing elderly participants if its inclusion criteria include *any* adults at age 65 or over, encompassing trials with only elderly participants or both elderly and non-elderly participants. We estimate equation (1) using the number of trials in each category as the outcome and a binary measure of *MedicareShare*.¹⁰

Panel A of Figure 2 shows no significant difference in trial enrollment patterns across high and low Medicare share diseases prior to 2000. However, after the memorandum, we observe an immediate increase in the number of clinical trials whose recruitment criteria include elderly participants. This increase remains large, positive, and statistically significant through 2005.¹¹ We find no change in the number of trials enrolling non-elderly participants. Appendix Figure A.4 confirms that these

¹⁰We focus on linear models using the count of clinical trials for four reasons. First, many ICD-9 code-quarter cells contain zero trials. Second, our data violate the key assumption of Poisson models—that the mean and variance are equal. Third, nonlinear models cannot accommodate the full set of controls we use, such as interactions between ICD-9 disease chapters and time. Fourth, using trial counts facilitates straightforward interpretation, allowing us to directly quantify how the memorandum translates into changes in the number of clinical trials. As a robustness check, Appendix Figure A.4 shows our results hold when using the log transformation of trial counts or a binary indicator for any trial initiation.

¹¹We stop our main analysis in 2005 to avoid issues with the introduction of Medicare Part D in 2006. Further, in 2005, several universities settled federal lawsuits regarding inaccurate Medicare billing for clinical trials. Consequently, trial sponsors were increasingly cautious about Medicare billing.

results hold under alternative specifications, including the log of trial counts and a binary indicator for any trial in a given disease and quarter.

Columns 1 and 2 in Panel A of Table 2 present the corresponding estimates from equation (2). These estimates are at the disease-year level. Column 2 shows that among high Medicare share diseases, the number of trials recruiting any elderly participants increases by 1.102 per year. There was no significant impact on trials recruiting non-elderly participants. Panel B shows similar results when using a continuous measure of Medicare share. Together, these results suggest that the memorandum provided sufficient incentive for firms to expand trial eligibility to increase opportunities for elderly representation.

4.3.2 Trial Enrollment Counts

We now examine whether the shifts in trial enrollment criteria translate into an increase in the *actual* number of elderly participants enrolled. These results are less precise, since we only observe enrollment information for 22 percent of trials overall and 11 percent of trials prior to 2000.

As motivation for our analysis, Table 3 presents summary statistics on participant age before and after the Medicare Memorandum. The mean age of enrollees in old (high Medicare share) diseases increases from 51.5 to 55.5 ($p = 0.03$), while the mean age of enrollees in young diseases only increases from 38.5 to 41 ($p = 0.50$). The share of elderly enrollees in old diseases increases from 7 percent to 20 percent ($p = 0.03$), compared to an increase from 0 percent to 4 percent in young diseases ($p = 0.19$). The number of enrollees is a noisier measure and is not statistically significant in any row. However, point estimates indicate that the average number of elderly enrollees increases for all diseases and in old diseases, while the number of enrollees under 65 decreased in both groups.

Panel B of Figure 2 illustrates the causal effect of the memorandum on the percentage of elderly enrollees by comparing diseases with high and low Medicare share. Before the memorandum, the coefficients are statistically insignificant. Beginning with trials that started in June 2000 after the memorandum, the share of elderly enrollees increases and remains elevated. Corresponding estimates in Table 2 provide supporting evidence: Column 3 of Panel A shows that the memorandum leads to an 18 percentage point relative increase in elderly enrollment shares among trials in high Medicare share diseases, although the estimates are not statistically significant ($p = 0.086$). In

Column 6 of Panel B, using a continuous measure of Medicare share, we find that a one unit increase in Medicare share (from 0 to 1) is associated with a statistically significant 79 percentage point increase in elderly enrollment share.

Taken together, these findings suggest that the memorandum reduces enrollment costs for elderly participants. In response, firms expand eligibility criteria and enroll more elderly individuals. The results highlight the role of financial incentives in shaping representation in product development.

4.3.3 Robustness Checks

In this section, we summarize additional robustness checks. First, we examine the potential role of concurrent shocks. Panel A of Figure 2 shows that the increase in trials whose recruitment criteria include elderly participants occurs several years before December 2003, when the Medicare Modernization Act (which introduced Medicare Part D) was signed into law. Medicare Part D was implemented in January 2006. To further show that our results are not driven by the introduction of Medicare Part D, Appendix Table A.1 confirms that our results hold when restricting the sample years to 1995 to 2003. Additionally, Appendix Table A.2 shows that our results are robust to including an additional control for the passage of Medicare Part D in 2003. Finally, we consider the potential confounding effect of the 1997 FDA Modernization Act, which established fast-track approval for drugs treating serious conditions, such as HIV and leukemia. To minimize the possible effects of this regulatory incentive, we exclude all ICD-9 codes linked to drugs receiving fast-track approval. The results, shown in Appendix Table A.3, are consistent with our main findings.

5 Impact on Innovation, Product Adoption, and Quality

5.1 Impact on the Rate of Clinical Trials

Having shown that the Medicare Memorandum increases elderly representation in clinical trials, we now examine whether this translates into broader changes in innovation, as measured by the rate of clinical trial activity. Descriptive evidence supports this hypothesis. Appendix Figure A.5 plots the total number of trials by Medicare share deciles, with diseases grouped into the top 10 percent, 10–20 percent, and so on. Trial counts increase across all disease categories prior to 2000,

likely reflecting improvements in trial registration and reporting. However, after 2000, the increase is disproportionately larger for diseases with a higher share of elderly diagnoses.

Figure 3 presents event study estimates showing an immediate and sustained increase in the number of trials in diseases with high versus low Medicare shares following the memorandum. Column 1 of Panel A in Table 4 shows that diseases with a high Medicare share experience an annual increase of 1.114 clinical trials relative to low Medicare share diseases. To further characterize these effects, Column 2 uses log-transformed outcomes and finds that the memorandum leads to a 24 percent increase in the number of clinical trials ($\approx e^{0.213} - 1$). Column 3 uses a binary outcome and shows a 29 percent increase in the likelihood of any trial in a given disease-year ($\approx \frac{0.0707}{0.248}$). Panel B replaces the binary Medicare share measure with a continuous one and finds similar results.

The relatively rapid increase in trials is consistent with the idea that firms were testing previously developed drugs, such as those repurposed for new indications and taken “off the shelf.” Table 5 supports this interpretation. The increase in total Phase II and III trials (corresponding to Column 1 of Table 4) is largely driven by drugs that had already undergone Phase II or III testing, rather than by drugs entering these phases for the first time. When examining trends by trial phase, we find that the immediate increase is especially pronounced for Phase II trials in old diseases, consistent with an “off the shelf” model (see Appendix Figure A.6).

The effects of the memorandum are likely to be greater in diseases where the cost of participant enrollment is historically high. Appendix Table A.4 shows that the rate of clinical trials disproportionately increases in diseases associated with longer trial durations, which may involve higher routine and retention costs. We also find that diseases more commonly treated in inpatient settings, which may have more expensive trials, had larger effects than outpatient diseases, though the difference is not statistically significant. These results are consistent with the view that the memorandum effectively lowered trial costs in diseases where participant enrollment is more costly or logistically challenging.

5.1.1 Robustness Checks

We evaluate the robustness of our results on the rate of clinical trial activity. First, we examine the potential role of concurrent shocks. Figure 3 shows that the increase in trials in diseases with above-

median elderly diagnoses begins several years before 2003, when the Medicare Modernization Act (which introduced Medicare Part D) was signed into law. Second, our results hold after restricting the sample to trials initiated before December 2003 (Column 1 of Appendix Table A.5) and when directly controlling for the passage of Part D (Column 2 of Appendix Table A.5). Third, our results are robust to excluding diseases likely to be affected by the FDA Modernization Act of 1997 (Column 3 of Appendix Table A.5).

We show that our trial results are likely not driven by a global shift toward elderly drugs. Since the memorandum primarily lowered the cost of trials run in the U.S., we would expect a limited impact on trials conducted in Europe. Appendix Figure A.7 shows that the increase in elderly-focused innovation is muted in Europe. Innovation in other countries may be a complement or substitute for U.S.-based innovation, but we would expect any spillover effects to be smaller than the main effect of the memorandum.

Finally, we investigate the possibility that projections about the U.S. elderly population shifted around 2000, which might have shaped demand for pharmaceuticals by elderly patients (and ultimately, the overall rate of clinical trial activity). However, the 2000 census did not show large gains in the elderly population: From 1990 to 2000, the growth of the elderly population was 12.0 percent, compared to the total population growth of 13.2 percent.¹² In 2000, the baby boomers, defined as those born 1946-1964, were still aged 36-54 and projections about their age patterns did not change discretely.

5.2 Impact on Drug Utilization

Lowering the cost of representation in the product development process may influence product adoption through several channels. For example, by reducing the cost of including elderly participants, the memorandum leads to more trials with greater elderly representation. This may have increased both the quantity and quality of evidence for both elderly diseases and elderly patients. In turn, this may lead to greater and more appropriate prescribing (McKibbin, 2023; Azoulay, 2002). An alternate, more speculative channel is that representation itself may directly impact adoption. Clinical trial results and demographics feature in marketing materials (see Appendix Figure A.8 for an

¹²See https://www.census.gov/newsroom/releases/archives/2010_census/cb11-cn192.html

example) and may cause physicians to perceive that a drug is now more appropriate for an elderly patient.¹³

To examine whether increases in representation shapes product adoption, we compare drug utilization among elderly versus middle-aged individuals, before and after the memorandum:

$$y_{it} = \sum_t \beta_t \mathbf{1}\{year = t\} \times \mathbf{1}\{Elderly_{a(i)}\} + \mathbf{1}\{Elderly_{a(i)}\} + \delta_t + \eta_i + \epsilon_{it} \quad (3)$$

where y_{it} is the number of unique prescriptions among either old or young diseases for patient i in year t .¹⁴ Prescriptions are matched to ICD-9 codes and classified as treating old or young diseases as described in Section 3.4. Age categories are denoted by $a(i)$ and $\mathbf{1}\{Elderly_{a(i)}\}$ refers to whether the individual is aged 65-75, relative to a control group aged 50-60. The δ_t are calendar year fixed effects and η_i includes individual-level controls for gender, race and ethnicity. Standard errors are robust and all regressions use MEPS analytical weights.

Figure 4 shows that following the memorandum, we observe a substantial increase in the use of elderly-oriented drugs among elderly patients, compared with both drugs targeted towards younger diseases and drug utilization patterns among non-elderly patients. Before the memorandum, use of drugs for old diseases for elderly patients was slightly increasing relative to younger patients but was closely aligned with patterns for drugs in young diseases. After the memorandum, the use of drugs for old diseases by the elderly begins to increase and the difference between old and young diseases becomes statistically significant in 2002. Figure 4 includes all MEPS prescriptions linked to either old or young diseases. Table 6, Column 2 shows that elderly individuals use 0.444 more unique prescriptions for old diseases, a 23 percent increase compared to a baseline mean of 1.925. We find no significant change in prescriptions for young diseases (Column 1).

Elderly drug utilization rose by 2002, before Medicare Part D was signed or enacted. As a robustness check that our results are not driven by anticipatory effects of Medicare Part D, we restrict our sample to individuals who earn above 400 percent of the federal poverty line and have private health insurance (Columns 3 and 4). In 1999, 86 percent of this group already had drug

¹³Physicians may learn about new trial evidence from pharmaceutical representatives, publications, continuing medical education requirements, or medical guidelines.

¹⁴In some years, survey respondents are asked what date they began their medication, but most dates reported are January 1st of the survey year.

insurance.¹⁵ Among this group, elderly patients reported that prescriptions for old diseases increases by 0.398, closely aligning with our main estimates.

Previous studies have found that the implementation of Medicare Part D in 2006 also resulted in an increase in drug utilization. [Duggan and Morton \(2010\)](#) find that drugs most often used by Medicare recipients increased in utilization by 28 percent to 70 percent from 2003 to 2006, though most of their estimates are not statistically significant. [Zhang et al. \(2009\)](#) and [Lichtenberg and Sun \(2007\)](#) also find increases in utilization, but their data begin in 2004 and do not examine the early 2000s. While prescriptions rose modestly after 2003 and 2006, prescription patterns for elderly individuals taking drugs for old diseases had already increased by 2003. These patterns were missed by prior studies that started in 2003 or 2004.

5.2.1 Heterogeneity by Clinical Trial History of Drug

To more directly examine the relationship between increases in representation in trials and subsequent patterns of drug utilization, we leverage heterogeneity in the clinical trial history of each drug. We investigate whether increases in drug utilization occur among drugs newly tested in trials after the Medicare memorandum. We define a treatment group consisting of the subsample of drugs newly tested in clinical trials between 2000 and 2005, excluding any drugs with trials in the pre-policy period (1990–1999).¹⁶ As a control group, we include drugs with no new trials at any point during the 1990–2005 period.¹⁷

Figure 5, Panel A, shows that drugs for old diseases newly tested in clinical trials from 2000–2005 exhibit no change in prescriptions during the pre-period, but a clear increase beginning in 2002–2003. In contrast, prescription growth for newly tested drugs in young diseases is smaller and occurs only after the implementation of Medicare Part D in 2006. Panel B displays trends for the control group: drugs not tested in clinical trials between 1990 and 2005. In this group, prescription rates for both old and young diseases remain flat until the onset of Medicare Part D.

¹⁵We cannot restrict to individuals with drug insurance as this question was not asked from 1996–1998.

¹⁶We treat a molecule as tested if any of its associated names (e.g., Zoloft, sertraline, sertraline hydrochloride) appear in a clinical trial.

¹⁷The third and omitted category consists of drugs tested before the policy (1990–1999), for which post-policy changes in prescriptions may reflect the effect of older trials or new ones. The most common drugs in this category are atorvastatin (Lipitor), lisinopril, and atenolol.

Table 6 reports corresponding estimates, restricting the sample to 1995–2005 to avoid confounding from Medicare Part D. Drugs newly tested between 2000 and 2005 show a statistically significant increase in prescriptions for both old and young diseases, but the effect for old diseases is much larger in absolute and percentage terms. For newly tested drugs in old diseases, prescriptions rise by 0.172 per person, a 27 percent increase relative to the pre-period mean of 0.634, and are slightly higher than the average effect across all drugs of 23 percent. In contrast, we find no change in prescriptions for drugs never tested in clinical trials.¹⁸

Consistent with our finding that increases are larger among trials testing new uses of previously tested drugs (Table 5), the drugs in the treatment group (i.e., those newly tested in 2000–2005) are almost all previously approved compounds undergoing additional trials. For elderly patients, the most prescribed drug in this group for the elderly is levothyroxine (Synthroid), which has been widely prescribed since the 1950s and approved as a “new drug” after additional testing and regulatory review in 2002. The second is metoprolol (Toprol), approved in 1978 but included in head-to-head trials against carvedilol, a newer patented alternative. The third is simvastatin (Zocor), approved in 1991 and studied in several cardiovascular trials in the early 2000s. In each case, prescriptions rose following the new trials. Of the ten most common drugs in this group, only albuterol was a new molecule approved after 2000. Drugs in the control group (i.e., not tested in clinical trials from 1990–2005) are typically long-established therapies. The most common include furosemide (approved in 1964) for hypertension, hydrochlorothiazide (1959), also for hypertension, and glipizide (1984) for diabetes.

To provide further support for the connection between prescriptions and trials, we examine the relationship between changes in trials and prescription utilization at the disease level. Appendix Figure A.9 plots the change in elderly prescriptions against the change in trials by ICD-9 code. Each point is an ICD-9 code. The relationship is positive and statistically significant (slope = 0.017, $p < 0.001$). Controlling for the Medicare share of each ICD-9 code yields a similar estimate (slope = 0.016, $p < 0.01$), indicating the relationship is not driven solely by the share of Medicare

¹⁸The increase in prescriptions for newly tested drugs cannot fully account for the overall effect, since we find large increases in prescriptions also for the omitted group, drugs that were tested from 1990–1999, which were usually also in trials in the 2000s. Unfortunately, we cannot isolate whether these increases are driven by pre- or post-memorandum trials.

recipients in each disease. These results suggest that increases in product adoption are concentrated in diseases with greater changes in innovative activity after the memorandum.

5.3 Impact on Quality

The previous analyses showed that diseases with increased representation and innovation were also associated with greater product adoption. As discussed in Section 5.2, this may occur because increased elderly representation and an increase in clinical trials may improve the level and quality of trial evidence. In turn, this can lead to increased and more appropriate prescribing. To examine changes in product quality (defined as the safety and effectiveness of the drug for its intended user group), we examine two proxies: the precision of trial results, and post-market adverse events.

First, by increasing elderly enrollment, the memorandum may have improved the precision of clinical trial results. Our analysis, which spans 1995-2005, covers only a subset of trials with reported results (mandatory results reporting to ClinicalTrials.gov began in 2007). Despite these data limitations, Panel A of Table 7 shows that following the memorandum, there is a greater increase in trials with precise estimates ($p < 0.05$) relative to trials without statistically significant outcomes ($p \geq 0.05$). However, the difference between these two measures are not statistically significant.

Second, we analyze post-market adverse events using the FDA’s Adverse Event Reporting System (AERS), which monitors drug safety after approval. We construct two measures: the total number of adverse events and the rate of adverse events per 1,000 unique MEPS prescriptions, as detailed in Appendix Section A.2. We then estimate changes in these outcomes for elderly users in old diseases, relative to younger users and young diseases. For each drug–age–year–gender cell, we regress

$$y_{d,a,g,t} = \beta_t Post_t Elderly_a + Elderly_a + \delta_{j(d)} + \eta_g + \epsilon_{d,a,g,t} \quad (4)$$

where $y_{d,a,g,t}$ is the adverse event count or rate for each drug d in age group a and gender g in year t . We run these separately for drugs for either old or young diseases as described in Section 4. $Post_t$ is an indicator for after the memorandum. $Elderly_a$ refers to whether the age is 65-75,

relative to a control category aged 50-60. $\delta_{j(d)}$ are indicators for the ICD-9 code j for each drug. η_g refer to person-level controls for gender. Standard errors are robust and we include analytical weights of the number of MEPS prescriptions in each cell.

Columns 3 and 4 of Table 7 show that adverse events decrease among the elderly in old diseases and increase in young diseases, though the difference is not statistically significant. Because adverse events are reported as counts, more common drugs naturally generate more reports. Columns 5 and 6 address this by analyzing adverse events per 1,000 prescriptions. Among the elderly in old diseases, the adverse event rate falls by 0.13 per 1,000 prescriptions after 2000, an 8 percent decline relative to a baseline adverse event rate of 1.5. The adverse event rate in other age and disease groups shows no change or even increases and the difference between the adverse event rate for old and young diseases has a p -value of 0.09. While adverse event data are subject to reporting bias and rely on MEPS-based estimates of prescription users, the findings are consistent with the view that increased representation in R&D improves both trial precision and post-market safety.

6 Discussion

The Medicare Memorandum reduced financial barriers to enrolling elderly participants in clinical trials. Routine costs, which include clinical procedures and labs, account for 25-50 percent of the direct costs of a clinical trial. In addition, firms benefited from lower patient recruitment costs due to the patient awareness component of the memorandum (Sertkaya et al., 2016).

This memorandum addressed several market failures. First, innovation has positive externalities, which is one reason why the government might increase funding for clinical trials. Second, the memorandum also included a patient awareness component, which likely reduced financial uncertainty and increased elderly enrollment. Third, the breadth of this memorandum likely led to a general equilibrium increase in elderly participation beyond what a single firm could generate. Such increases in trial representation could result in social benefits due to potential knowledge spillovers.

Finally, firms typically have limited incentives to conduct trials for new uses for existing drugs, as their returns diminish once patent exclusivity ends (Eisenberg, 2005). While pharmaceutical companies could have subsidized participation directly (rather than relying on government subsi-

dies), their investment typically leads to greater returns for novel, potential blockbuster drugs.¹⁹ In contrast, subsidizing participation for trials testing new approvals for existing drugs typically leads to more limited financial benefits.

Consistent with this, we find that the memorandum primarily led to additional trials testing new uses for existing drugs (Table 5). Panel A of Appendix Figure A.10 further shows no increase in the number of newly approved drugs in old diseases after the policy. In contrast, Panel B shows an increase in the number of drugs suspended, withdrawn, or discontinued per disease-year, although this change is not statistically significant. This pattern is consistent with firms investing in trials with higher ex-ante risk of termination. Some of these trials may fail, but others may produce new evidence of efficacy (e.g., for alternative indications, secondary outcomes, or subpopulations). In terms of magnitudes, Table 4 shows an average annual increase of 1.1 clinical trials in high Medicare share diseases, compared to just 0.2 additional drugs that result in suspensions, withdrawals, or discontinuations. These findings suggest that most new trials generate positive evidence that increases prescribing without leading to new approvals, while a smaller share yields negative results. Regardless of the trial outcomes, such evidence may still be used to increase utilization as off-label prescribing is permitted.

Collectively, our findings suggest that the reduction of financial barriers driven by Medicare Memorandum and government subsidization of clinical trials, more generally, increases representation, innovation, and prescriptions. While firms experience financial benefits in terms of increased utilization, we expect they are somewhat limited, as additional trials are primarily used to generate additional information on existing drugs or test medications with a higher ex-ante risk of termination, rather than to directly support regulatory approval for novel drugs.

7 Conclusion

This paper provides new causal evidence on the causes and consequences of representation in product development. We leverage a policy that lowered financial barriers through expanding insurance coverage for elderly trial participants and find that this led trial sponsors to broaden enrollment criteria. We also provide evidence that enrollment of elderly individuals increases. These shifts in

¹⁹As mentioned in Section 2.2, about one-third of all industry trials were fully funded and covered all routine costs without billing Medicare before the memorandum.

representation corresponded with a 24 percent increase in the number of clinical trials for diseases common among the elderly and a significant rise in drug utilization by elderly patients. Additional evidence from trial outcomes and adverse event reports suggests that improvements in actual and perceived drug quality may be an important mechanism linking representation to increased product adoption. Together, our findings highlight that reducing financial constraints can increase representation and meaningfully shape innovation, product adoption, and quality.

Policymakers and researchers have called for governments to address the persistent underrepresentation of specific demographic groups—specifically, the elderly, women, and Black individuals—in clinical trials relative to the prevalence of the diseases affecting these groups. Our findings suggest that policy efforts to increase representation within product development may generate larger impacts by reducing financial constraints to participation (Takvorian, Guerra and Schpero, 2021). For firms, the relationship between representation and product adoption highlights the benefits of increasing representation in the product development process. While direct subsidization of specific patient subgroups may not be worthwhile for all trials (e.g., trials testing new uses of approved drugs), firms may benefit from low cost, scalable investments that reduce the cost of participation. These interventions can include collaborations with other firms to share recruitment infrastructure or adopting digital trial platforms that make participation more accessible (Inan et al., 2020).

Our findings highlight that greater representation can spur subsequent innovation. In our setting, subsidizing elderly representation benefited firms. However, greater representation may also limit subsequent innovation. Policies that mandate increases in representation without a sufficient change in financial incentives may impose costs and delay product approvals. This may be particularly harmful in markets where demographic representation is less salient to target users or where expediting market entry is especially important. Future work should explore the policy and market conditions under which efforts to increase representation ultimately spur rather than constrain innovation. Second, we provide suggestive evidence that increased representation improves trial precision and quality. However, data limitations prevent a full analysis of how trial outcomes, such as efficacy or survival improvements, change with representation. A more comprehensive evaluation of how representation shapes product outcomes could offer important insights into the broader impact of increasing representation in product development.

References

- Aaron, Henry J, and Hellen Gelbrand.** 2000. “Extending Medicare reimbursement in clinical trials.” *National Academies Press*.
- Acemoglu, Daron, David Cutler, Amy Finkelstein, and Joshua Linn.** 2006. “Did medicare induce pharmaceutical innovation?” *American Economic Review*, 96(2): 103–107.
- Agha, Leila, Soomi Kim, and Danielle Li.** 2022. “Insurance design and pharmaceutical innovation.” *American Economic Review: Insights*, 4(2): 191–208.
- Alpert, Abby.** 2016. “The anticipatory effects of Medicare Part D on drug utilization.” *Journal of Health Economics*, 49: 28–45.
- Alsan, Marcella, and Marianne Wanamaker.** 2017. “Tuskegee and the Health of Black Men*.” *The Quarterly Journal of Economics*, 133(1): 407–455.
- Alsan, Marcella, Maya Durvasula, Harsh Gupta, Joshua Schwartzstein, and Heidi Williams.** 2024. “Representation and Extrapolation: Evidence from Clinical Trials.” *Quarterly Journal of Economics*, 139(1): 575–635.
- Alsan, Marcella, Romaine A. Campbell, Lukas Leister, and Ayotomiwa Ojo.** 2025. “Investigator racial diversity and clinical trial participation.” *Journal of Health Economics*, 100: 102968.
- Azoulay, Pierre.** 2002. “Do Pharmaceutical Sales Respond to Scientific Evidence?” *Journal of Economics & Management Strategy*, 11(4): 551–594.
- Blume-Kohout, Margaret E, and Neeraj Sood.** 2013. “Market size and innovation: Effects of Medicare Part D on pharmaceutical research and development.” *Journal of Public Economics*, 97: 327–336.
- Camuffo, Arnaldo, Alessandro Cordova, Alfonso Gambardella, and Chiara Spina.** 2020. “A scientific approach to entrepreneurial decision making: Evidence from a randomized control trial.” *Management Science*, 66(2): 564–586.

- Cao, Ruiqing, Rembrand Koning, and Ramana Nanda. 2023. "Sampling Bias in Entrepreneurial Experiments." *Management Science*.
- Cortellis Clinical Trials Intelligence Database. 1995-2005. *Clarivate Analytics*. <https://clarivate.com/products/cortellis-family/>.
- Cox, KM, S Goel, RL O'connell, M Boyer, PJ Beale, RJ Simes, and MR Stockler. 2011. "Randomized cross-over trial comparing inpatient and outpatient administration of high-dose cisplatin." *Internal Medicine Journal*, 41(2): 172–178.
- DiMasi, Joseph A, Henry G Grabowski, and Ronald W Hansen. 2016. "Innovation in the pharmaceutical industry: new estimates of R&D costs." *Journal of Health Economics*, 47: 20–33.
- DiMasi, Joseph A, Ronald W Hansen, and Henry G Grabowski. 2003. "The price of innovation: new estimates of drug development costs." *Journal of Health Economics*, 22(2): 151–185.
- Duggan, Mark, and Fiona Scott Morton. 2010. "The effect of Medicare Part D on pharmaceutical prices and utilization." *American Economic Review*, 100(1): 590–607.
- Einiö, Elias, Josh Feng, and Xavier Jaravel. 2019. "Social push and the direction of innovation." Available at SSRN: <https://ssrn.com/abstract=3383703>.
- Eisenberg, Rebecca S. 2005. "The problem of new uses." *Yale J. Health Pol'y L. & Ethics*, 5: 717.
- Ewens, Michael, and Richard R Townsend. 2020. "Are early stage investors biased against women?" *Journal of Financial Economics*, 135(3): 653–677.
- FDA Adverse Events Reporting System. 1969-2006. Obtained via FOIA request through Investigative Reporters and Editors <https://www.ire.org/>.
- Friedrichs, Anke, Maren Spies, Martin Härter, and Angela Buchholz. 2016. "Patient preferences and shared decision making in the treatment of substance use disorders: a systematic review of the literature." *PloS one*, 11(1): e0145817.

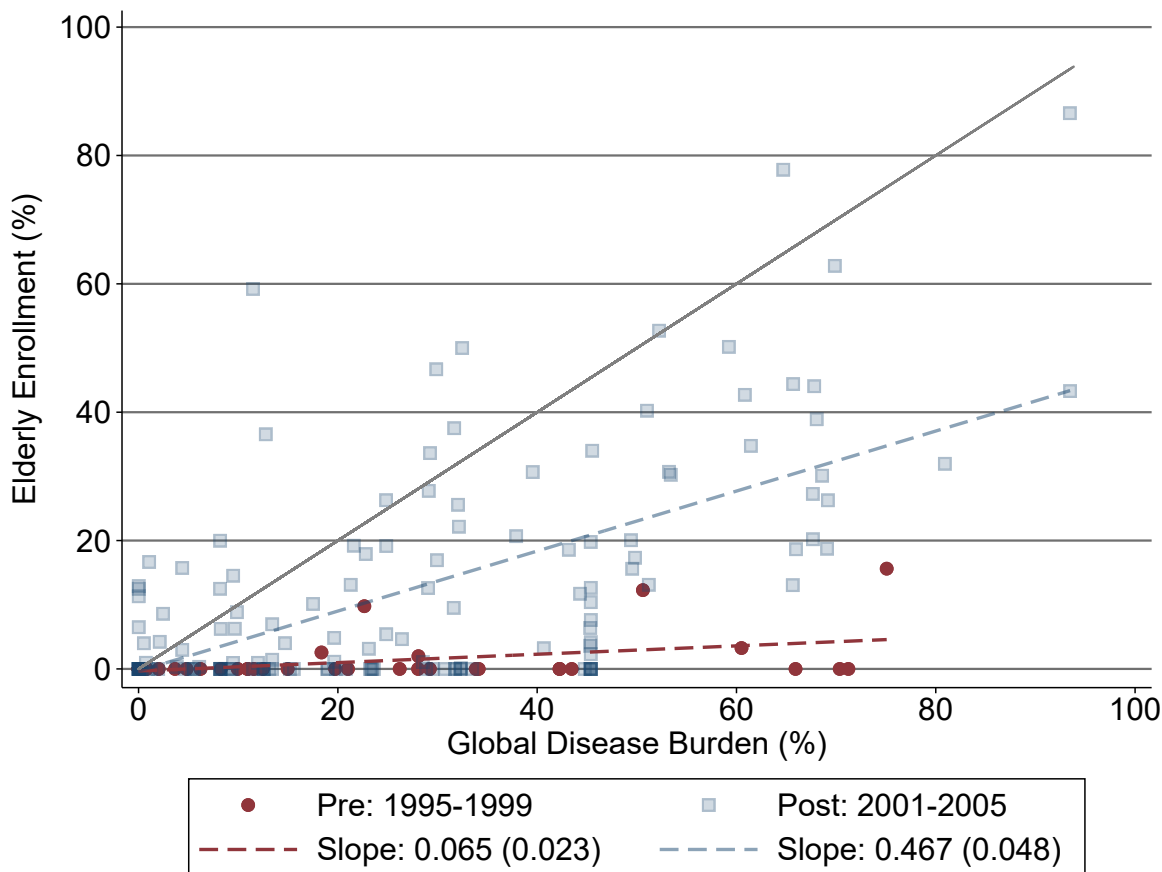
- Global Burden of Disease. 2019. <https://www.healthdata.org/research-analysis/gbd-data>.
- Gompers, Paul A, and Sophie Q Wang. 2017. “Diversity in innovation.” National Bureau of Economic Research, Working Paper 23082.
- Gupta, Harsh. 2022. “Do Female Researchers Increase Female Enrollment in Clinical Trials?” *Working Paper*.
- Guzman, Jorge, and Aleksandra Olenka Kacperczyk. 2019. “Gender gap in entrepreneurship.” *Research Policy*, 48(7): 1666–1680.
- Hebert, Camille. 2025. “Gender Stereotypes and Entrepreneur Financing.” *The Review of Financial Studies*.
- Hegde, Deepak, and Justin Tumlinson. 2014. “Does social proximity enhance business partnerships? Theory and evidence from ethnicity’s role in US venture capital.” *Management Science*, 60(9): 2355–2380.
- Herrera, Angelica P., Shedra Amy Snipes, Denae W. King, Isabel Torres-Vigil, Daniel S. Goldberg, and Armin D. Weinberg. 2010. “Disparate Inclusion of Older Adults in Clinical Trials: Priorities and Opportunities for Policy and Practice Change.” *American Journal of Public Health*, 100(S1): S105–S112.
- Howell, Sabrina T, and Ramana Nanda. 2024. “Networking frictions in venture capital, and the gender gap in entrepreneurship.” *Journal of Financial and Quantitative Analysis*, 59(6): 2733–2761.
- Hutchins, Laura F, Joseph M Unger, John J Crowley, Charles A Coltman Jr, and Kathy S Albain. 1999. “Underrepresentation of patients 65 years of age or older in cancer-treatment trials.” *New England Journal of Medicine*, 341(27): 2061–2067.
- Inan, Omer T, P Tenaerts, Sheila A Prindiville, HR Reynolds, DS Dizon, K Cooper-Arnold, M Turakhia, Mark J Pletcher, Kenzie L Preston, Harlan M Krumholz, et al. 2020. “Digitizing clinical trials.” *NPJ digital medicine*, 3(1): 1–7.

- Kanze, Dana, Laura Huang, Mark A Conley, and E Tory Higgins.** 2018. “We ask men to win and women not to lose: Closing the gender gap in startup funding.” *Academy of management journal*, 61(2): 586–614.
- Kerr, William R, Ramana Nanda, and Matthew Rhodes-Kropf.** 2014. “Entrepreneurship as experimentation.” *Journal of Economic Perspectives*, 28(3): 25–48.
- Koning, Rembrand, Sampsa Samila, and John-Paul Ferguson.** 2021. “Who do we invent for? Patents by women focus more on women’s health, but few women get to invent.” *Science*, 372(6548): 1345–1348.
- Koning, Rembrand, Sharique Hasan, and Aaron Chatterji.** 2022. “Experimentation and start-up performance: Evidence from A/B testing.” *Management Science*, 68(9): 6434–6453.
- Lara, Primo N., Roger Higdon, Nelson Lim, Karen Kwan, Michael Tanaka, Derick H.M. Lau, Ted Wun, Jeanna Welborn, Frederick J. Meyers, Scott Christensen, Robert O’Donnell, Carol Richman, Sidney A. Scudder, Joseph Tuscano, David R. Gandara, and Kit S. Lam.** 2001. “Prospective Evaluation of Cancer Clinical Trial Accrual Patterns: Identifying Potential Barriers to Enrollment.” *Journal of Clinical Oncology*, 19: 1728–1733.
- Lau, SW Johnny, Yue Huang, Julie Hsieh, Shenggang Wang, Qi Liu, Patricia W Slatum, Janice B Schwartz, Shiew-Mei Huang, and Robert Temple.** 2022. “Participation of older adults in clinical trials for new drug applications and biologics license applications from 2010 through 2019.” *JAMA Network Open*, 5(10): e2236149–e2236149.
- Lichtenberg, Frank R., and Shawn X. Sun.** 2007. “The Impact Of Medicare Part D On Prescription Drug Use By The Elderly.” *Health Affairs*, 26(6).
- Luo, Hong, and Laurina Zhang.** 2025. “Gender Inequality and Product Choice: Evidence from #MeToo.”
- McKibbin, Rebecca.** 2023. “The effect of RCTs on drug demand: Evidence from off-label cancer drugs.” *Journal of Health Economics*, 90.

- Medical Expenditure Panel Survey.** 1996-2010. *Agency for Healthcare Research and Quality*.
https://meps.ahrq.gov/mepsweb/data_stats/data_overview.jsp.
- Michelman, Valerie, and Lucy Msall.** 2022. “Sex, Drugs, and R&D: Missing Innovation from Regulating Female Enrollment in Clinical Trials.” *Working Paper*.
- Miller, Douglas L.** 2023. “An Introductory Guide to Event Study Models.” *Journal of Economic Perspectives*, 37(2): 203–30.
- Nielsen, Mathias Wullum, Jens Peter Anderson, Londa Schiebinger, and Jesper W. Schneider.** 2017. “One and a half million medical papers reveal a link between author gender and attention to gender and sex analysis.” *Nature Human Behavior*, 1: 791–796.
- Pear, Robert.** 2000. “Clinton to Order Medicare to Pay New Costs.” *The New York Times*, A p.24.
- Sertkaya, Aylin, Hui-Hsing Wong, Amber Jessup, and Trinidad Beleche.** 2016. “Key cost drivers of pharmaceutical clinical trials in the United States.” *Clinical Trials*, 13(2): 117–126.
- Takvorian, Samuel U., Carmen E. Guerra, and William L. Schpero.** 2021. “A Hidden Opportunity — Medicaid’s Role in Supporting Equitable Access to Clinical Trials.” *New England Journal of Medicine*, 384(21): 1975–1978.
- Truffa, Francesca, and Ashley Wong.** 2025. “Undergraduate Gender Diversity and the Direction of Scientific Research.” *American Economic Review*, 115(7): 2414–48.
- U.S. Census Bureau.** 2023. “Census Bureau Releases New 2020 Census Data on Age, Sex, Race, Hispanic Origin, Households and Housing.” <https://www.census.gov/newsroom/press-releases/2023/2020-census-demographic-profile-and-dhc.html>.
- Zhang, Yuting, Julie M. Donohue, Judith R. Lave, Gerald O’Donnell, and Joseph P. Newhouse.** 2009. “The Effect of Medicare Part D on Drug and Medical Spending.” *New England Journal of Medicine*, 361(1): 52–61.

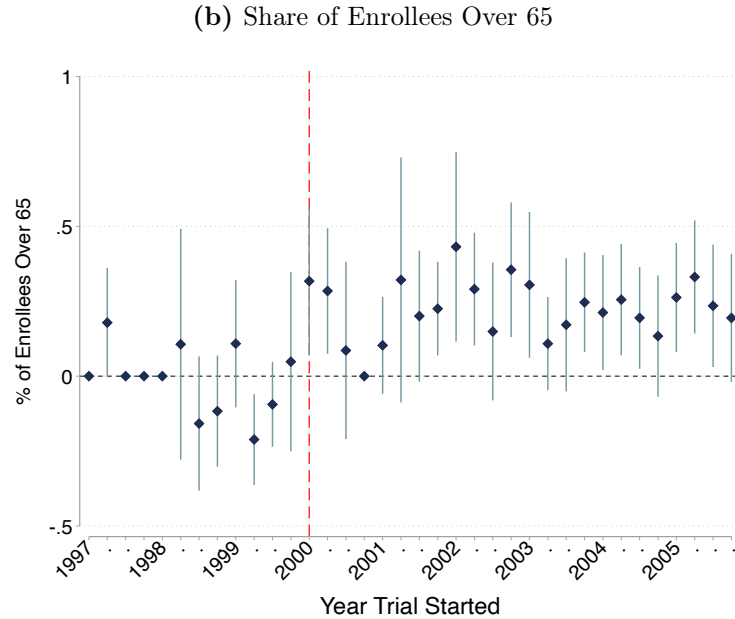
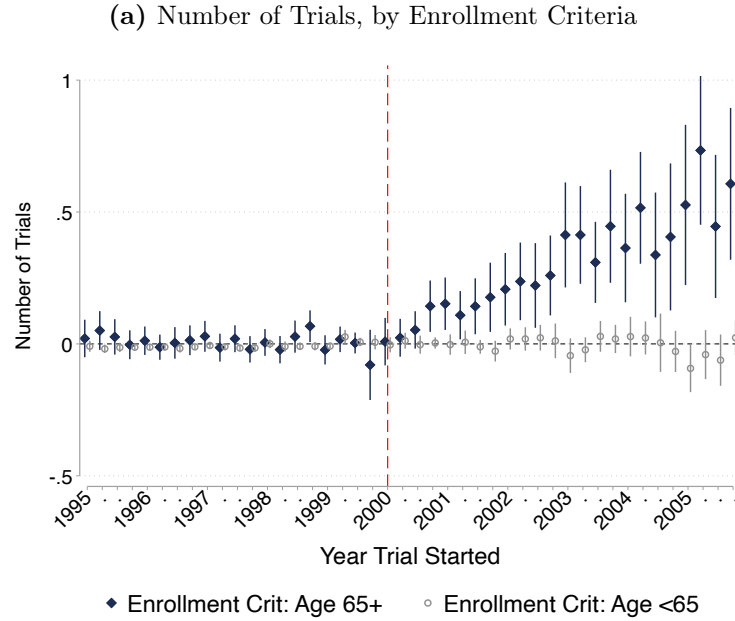
Figures and Tables

Figure 1: DISPARITIES IN ELDERLY ENROLLMENT OVER TIME



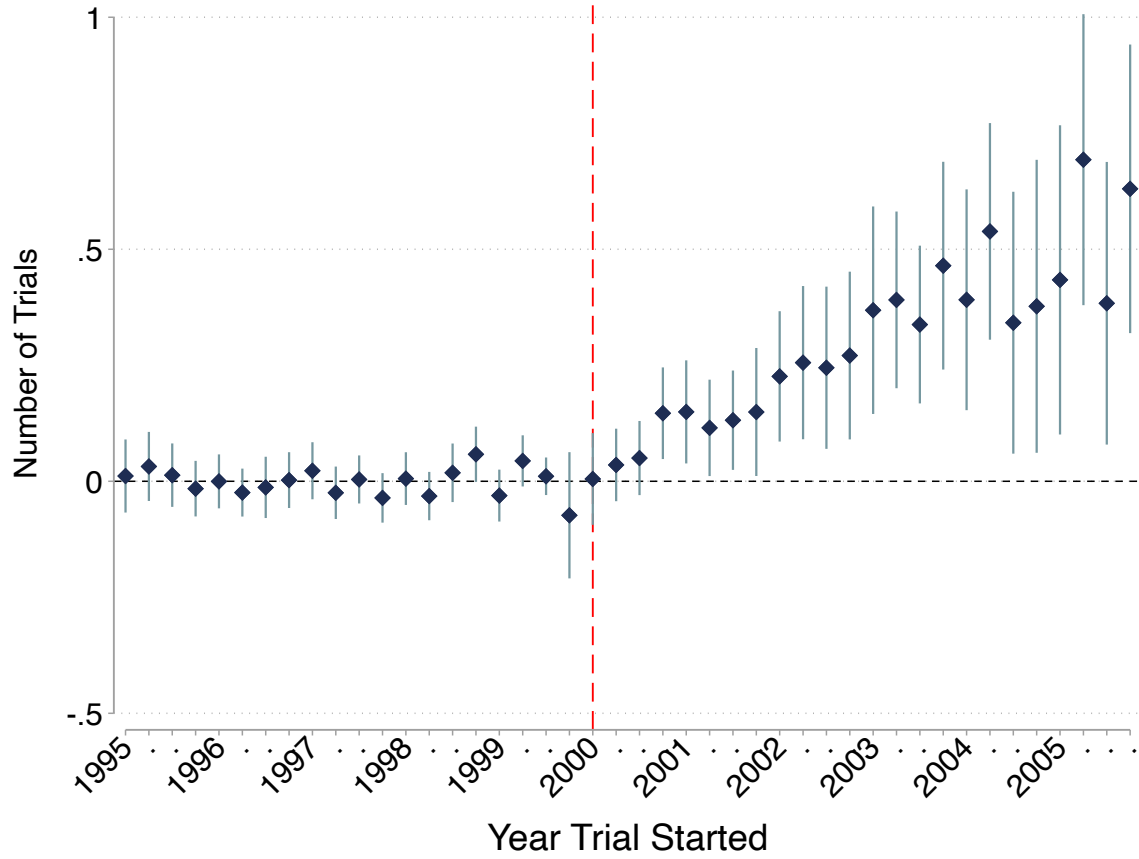
NOTE: Each point represents an ICD-9 code, with the x-axis showing the share of the disease burden that is among the elderly (aged over 65) and the y-axis presenting the average share of elderly participants in clinical trials for those diseases in our sample. The 45-degree line represents perfect parity between disease burden and trial enrollment. Points below this line indicate diseases where the elderly disease burden exceeds elderly trial enrollment. The darker dots average across trials that begin from 1995-1999 while the lighter squares average across 2001-2005. These two lines have statistically different slopes with $p < 0.001$.

Figure 2: IMPACT OF MEDICARE REIMBURSEMENT EXPANSION ON ENROLLMENT



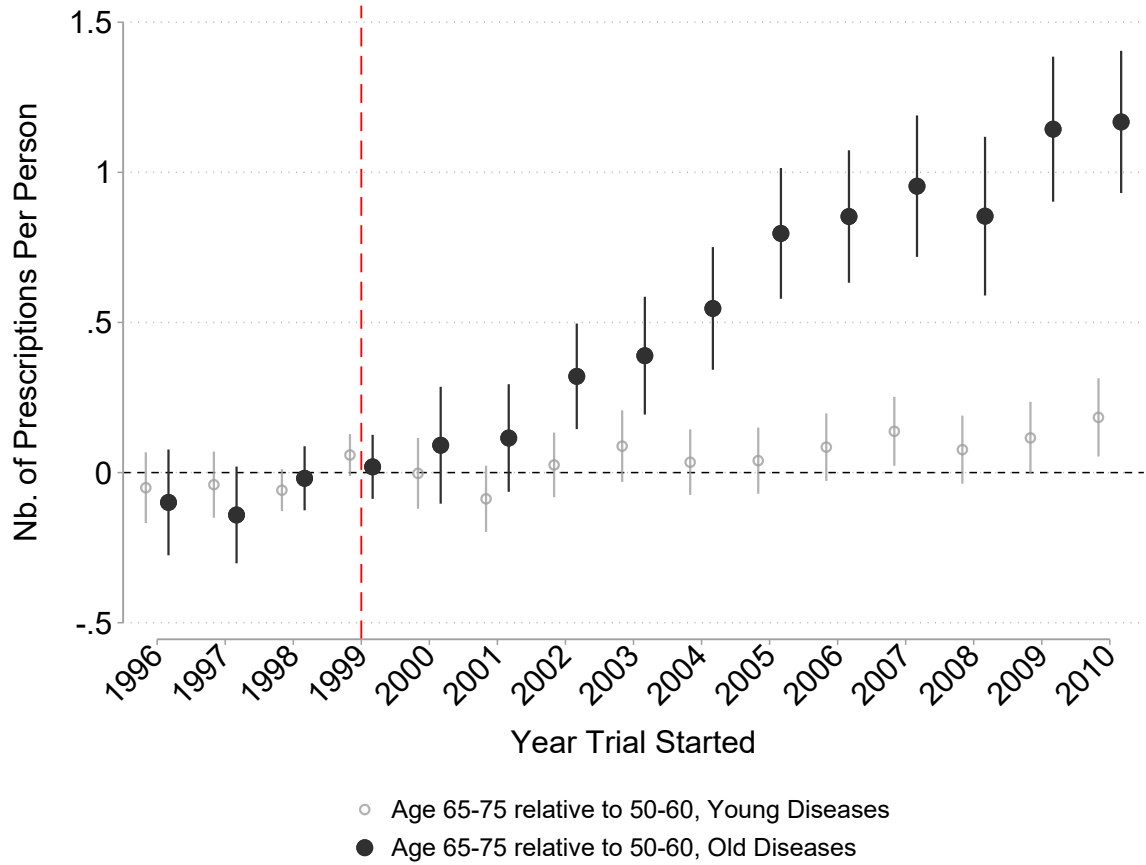
NOTE: This figure presents event study coefficients from the estimation of equation (1). Observations are at the disease-quarter level. We plot the coefficients on the interaction between quarter and an indicator for whether a disease had an above median share of diagnoses among the elderly from 1996-1999. Panel A presents estimates from two different regressions. In one the outcome is the number of trials that include elderly participants. In the other the outcome is the number of trials that do not include elderly participants in their enrollment criteria. All trials are in one of these two categories. In Panel B the outcome is the share of enrollees that are older than 65. There are only three trials with enrollment information before 1997, so we drop years before 1997. Disease-year combinations with no enrollment information are considered missing. Solid vertical lines are 95 percent confidence intervals. The dotted vertical line represents the quarter before the memorandum.

Figure 3: IMPACT OF MEDICARE MEMORANDUM ON NUMBER OF TRIALS



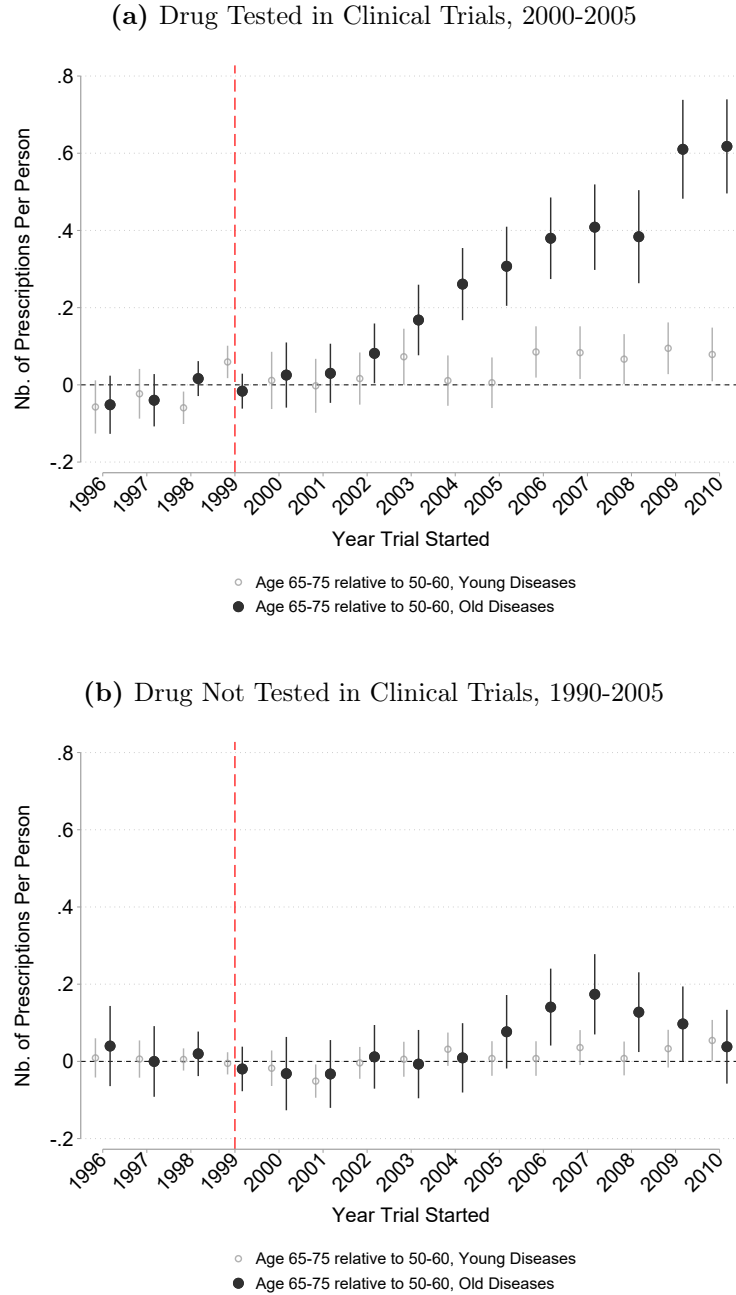
NOTE: This figure presents event study coefficients from the estimation of equation (1). Observations are at the disease-quarter level. We plot the coefficients on the interaction between quarter and an indicator for whether a disease had an above median share of diagnoses among the elderly from 1996-1999. The outcome is the number of trials in a quarter. Solid vertical lines are 95 percent confidence intervals. The dotted vertical line represents the quarter before the memorandum.

Figure 4: PRESCRIPTIONS BY AGE AND DISEASE CATEGORY, ALL DRUGS



NOTE: This figure presents event study coefficients from the estimation of equation (3). Observations are at the patient-year level. We plot the coefficients on the interaction between an indicator for age 65-75 (relative to 50-60) and year. The outcome is the number of unique prescriptions per person in diseases with an above median share of diagnoses among the elderly (“Old diseases”), or in diseases with a below median share of diagnoses among the elderly (“Young Diseases”). Solid vertical lines are 95 percent confidence intervals. The dotted vertical line represents the quarter before the memorandum.

Figure 5: PRESCRIPTIONS BY CLINICAL TRIAL HISTORY OF DRUG



NOTE: This figure is similar to Figure 4 but the outcome is the number of prescriptions for a subset of drugs. In Panel A, we consider prescriptions among drugs that were not tested in clinical trials from 1990 through May of 2000 but were tested in at least one clinical trial after the memorandum from June 2000 through 2005. In Panel B, the outcome includes prescriptions among drugs that were not tested in clinical trials from 1990-2005. Solid vertical lines are 95 percent confidence intervals. The dotted vertical line represents the quarter before the memorandum.

Table 1: SUMMARY STATISTICS

	Count (1)	Mean (2)	Median (3)	SD (4)	Min (5)	Max (6)
<i>Panel A. Disease-Quarter Level</i>						
Medicare Share	4,290	0.32	0.27	0.26	0	1
# Trials: Enrolling Only Age < 65	4,290	0.15	0.00	0.63	0	10
# Trials: Enrolling Any Age $\geq$ 65	4,290	0.93	0.00	3.32	0	56
Share of Enrolled Participants Age $\geq$ 65	478	0.13	0.00	0.22	0	1
# Trials: Phase II and III	4,290	1.09	0.00	3.51	0	58
<i>Panel B. Patient-Year Level</i>						
# Prescriptions						
Patients Age 50-60						
Higher Medicare Share Diseases	34,095	1.67	1.00	2.36	0	25
Low Medicare Share Diseases	34,095	1.04	0.00	1.66	0	24
Patients Age 65-75						
Higher Medicare Share Diseases	19,444	2.85	2.00	2.91	0	24
Low Medicare Share Diseases	19,444	1.05	0.00	1.56	0	20

NOTE: This table presents summary statistics for outcomes at the disease-year level (Panel A) and the patient-year level (Panel B). Medicare share is the share of diagnoses in that disease among Medicare patients. “# Trials: Enrolling Only Age <65” includes the number of trials with enrollment criteria that do not include elderly individuals. “# Trials: Enrolling Any Age $\geq$ 65” includes the number of trials with enrollment criteria that include elderly individuals. “Share of Enrolled Participants Age $\geq$ 65 has fewer observations because it is limited to disease-quarter pairs that have at least one trial with reported enrollment information. “# of Prescriptions” refers to the number of unique prescriptions per person in that disease category in the MEPS. Both panels are restricted to years 1995-2005.

Table 2: IMPACT ON ENROLLMENT IN CLINICAL TRIALS

<i>A. Binary Medicare Share</i>			
	# Trials Enrolling Only Age < 65 (1)	# Trials Enrolling Any Age ≥ 65 (2)	Share of Enrolled Participants Age ≥ 65 (3)
Post × $\mathbf{1}\{MedicareShare\}$	0.0119 (0.0548)	1.102*** (0.240)	0.183 (0.106)
Mean of Dep. Var.	0.148	0.941	0.129
Diff. Wald Test p -value	0.00		
Observations	4,290	4,290	409
Adjusted R^2	0.295	0.508	0.332
Year FEs	Yes	Yes	Yes
ICD9 FEs	Yes	Yes	Yes
ICD9 Category × Year Trends	Yes	Yes	Yes
<i>B. Continuous Medicare Share</i>			
	# Trials Enrolling Only Age < 65 (4)	# Trials Enrolling Any Age ≥ 65 (5)	Share of Enrolled Participants Age ≥ 65 (6)
Post × MedicareShare	-0.160 (0.0921)	1.935** (0.615)	0.789*** (0.173)
Mean of Dep. Var.	0.148	0.941	0.129
Diff. Wald Test p -value	0.00		
Observations	4,290	4,290	409
Adjusted R^2	0.296	0.507	0.351
Year FEs	Yes	Yes	Yes
ICD9 FEs	Yes	Yes	Yes
ICD9 Category × Year Trends	Yes	Yes	Yes

NOTE: Panel A presents coefficients from a disease-year level regression that regresses the outcome in each column on an indicator for after the Medicare Memorandum of 2000 (“Post”) interacted with an indicator for whether a disease has an above median share of diagnoses among the elderly (“ $\mathbf{1}\{MedicareShare\}$ ”). Panel B presents coefficients from a separate regression where “Post” is interacted with a continuous rather than binary measure of share of diagnoses among the elderly (“Medicare Share”). In both, we control for ICD-9 code fixed effects (which encompass “ $\mathbf{1}\{MedicareShare\}$ ” or “MedicareShare”), year fixed effects (which encompass the “Post” indicator) and the ICD-9 category by year trends. Columns 1 and 2 and separately, 4 and 5, are estimated using seemingly unrelated regressions, which permits a statistical comparison of the coefficients; Column 3 and 6 are estimated using OLS. The “Diff. Wald Test p -value” is the p -value comparing the coefficients in Columns 1 vs. 2 or Columns 4 vs 5. Singleton observations are dropped in fixed effects specifications, which accounts for the smaller number of observations than that shown in Table 1. Standard errors are clustered at the ICD-9 code level and reported in parentheses, with * $p < 0.05$; ** $p < 0.01$ and *** $p < 0.001$.

Table 3: CHANGES IN AGE OF ENROLLED TRIAL PARTICIPANTS

	Trials Starting Pre-2000			Trials Starting Post-2000			Diff of Means <i>p</i> -value
	Count (1)	Mean (2)	SD (3)	Count (4)	Mean (5)	SD (6)	
Mean Age							
All Diseases	88	48.3	17.2	772	51.7	15.8	0.05
Young Diseases	22	38.5	11.1	201	41.0	16.9	0.50
Old Diseases	66	51.5	17.8	571	55.5	13.5	0.03
% of Enrollees Over 65							
All Diseases	43	0.04	0.16	1,061	0.13	0.24	0.01
Young Diseases	20	0.00	0.01	430	0.04	0.13	0.19
Old Diseases	23	0.07	0.21	631	0.20	0.28	0.03
No. of Enrollees Over 65							
All Diseases	64	84	498	1,061	113	937	0.80
Young Diseases	28	0.04	0.19	430	19	191	0.61
Old Diseases	36	148	660	631	178	1,200	0.88
No. of Enrollees Under 65							
All Diseases	62	359	1,002	1,053	350	1,013	0.94
Young Diseases	28	359	938	427	533	1,443	0.53
Old Diseases	34	359	1,066	626	224	519	0.17

NOTE: This table shows the mean age, percentage over 65, and count of the enrolled trial participants over time. Observations are at the trial-level. “All Diseases” refers to the full sample of trials. Due to the limited number of observations from missing trial enrollment data, we have expanded the sample to include all trials initiated between 1990 and 2005, excluding those from the year 2000. As in the main analysis, we restrict to Phase II and Phase III interventional trials with private sector funding and at least one U.S. location. In instances where mean age was not available, median age was used instead. Column 7 shows *p*-values from a 2-sided t-test that compares Columns 2 and 5.

Table 4: IMPACT ON CLINICAL TRIAL ACTIVITY

<i>A. Binary Medicare Share</i>			
	# Trials (1)	Ln(# Trials) (2)	1(Any Trial) (3)
Post $\times$ $\mathbf{1}\{MedicareShare\}$	1.114*** (0.270)	0.213*** (0.0539)	0.0707* (0.0311)
Mean of Dep. Var.	1.089	0.336	0.248
Observations	4,290	4,290	4,290
Adjusted R^2	0.511	0.611	0.472
Year FEs	Yes	Yes	Yes
ICD9 FEs	Yes	Yes	Yes
ICD9 Category $\times$ Year Trends	Yes	Yes	Yes
<i>B. Continuous Medicare Share</i>			
	# Trials (4)	Ln(# Trials) (5)	1(Any Trial) (6)
Post $\times$ MedicareShare	1.776** (0.664)	0.394** (0.126)	0.153* (0.0642)
Mean of Dep. Var.	1.089	0.336	0.248
Observations	4,290	4,290	4,290
Adjusted R^2	0.509	0.610	0.472
Year FEs	Yes	Yes	Yes
ICD9 FEs	Yes	Yes	Yes
ICD9 Category $\times$ Year Trends	Yes	Yes	Yes

NOTE: Panel A presents coefficients from a disease-year level regression that regresses the outcome in each column on an indicator for after the Medicare Memorandum of 2000 (“Post”) interacted with an indicator for whether a disease has an above median share of diagnoses among the elderly (“ $\mathbf{1}\{MedicareShare\}$ ”). Panel B presents coefficients from a separate regression where “Post” is interacted with a continuous rather than binary measure of share of diagnoses among the elderly (“Medicare Share”). In both, we control for ICD-9 code fixed effects (which encompass “ $\mathbf{1}\{MedicareShare\}$ ” or “MedicareShare”), year fixed effects (which encompass the “Post” indicator) and the ICD-9 category by year trends. Coefficients are from OLS regressions. Standard errors are clustered at the ICD-9 code level and reported in parentheses, with * $p < 0.05$; ** $p < 0.01$ and *** $p < 0.001$.

**Table 5: IMPACT ON TRIALS TESTING NOVEL DRUGS
AND PREVIOUSLY TESTED DRUGS**

	Dependent Variable: # Trials	
	Testing Any Novel Drugs (1)	Testing Only Previously Tested Drugs (2)
$\text{Post} \times \mathbf{1}\{MedicareShare\}$	0.329*** (0.0778)	0.784*** (0.197)
Mean of dep. var.	0.310	0.778
Diff. Wald Test p -value	0.00	
Observations	4,290	4,290
Adjusted R^2	0.437	0.486
Year FEs	Yes	Yes
ICD9 FEs	Yes	Yes
ICD9 Category $\times$ Year Trends	Yes	Yes

NOTE: This table presents coefficients from regressing the outcome in each column on an indicator for after the Medicare Memorandum of 2000 (“Post”) interacted with an indicator for whether a disease has an above median share of diagnoses among the elderly (“ $\mathbf{1}\{MedicareShare\}$ ”). Observations are at the disease year level. The outcome in Column 1 is the number of trials that test drugs not previously investigated in any phase II or phase III trial prior to the focal trial. The outcome in Column 2 is the number of trials testing only drugs that have been previously evaluated in prior phase II or phase III trials before the focal trial. We control for ICD-9 code fixed effects (which encompass “ $\mathbf{1}\{MedicareShare\}$ ”), year fixed effects (which encompass the “Post” indicator) and ICD-9 category by year trends. The “Diff. Wald Test p -value” is the p -value comparing the $\text{Post} \times MedicareShare$ coefficient in Columns 1 vs. 2. Estimates come from seemingly unrelated regressions. Standard errors are clustered at the ICD-9 code level and reported in parentheses, with * $p < 0.05$; ** $p < 0.01$ and *** $p < 0.001$.

Table 6: IMPACT ON PRESCRIPTIONS

<i>A. All Drugs</i>					
	All Patients		High Income, Private Insurance Patients		
	# Prescriptions in Young Diseases (1)	# Prescriptions in Old Diseases (2)	# Prescriptions in Young Diseases (3)	# Prescriptions in Old Diseases (4)	
Post $\times \mathbf{1}\{Elderly\}$	0.0407 (0.0304)	0.444*** (0.0498)	-0.0209 (0.0541)	0.398*** (0.0821)	
Mean of Dep. Var.	0.966	1.925	0.992	1.581	
Diff. Wald Test p -value	0.00		0.00		
Observations	53,539	53,539	19,949	19,949	
Adjusted R^2	0.060	0.080	0.075	0.082	
Elderly FE	Yes	Yes	Yes	Yes	
Race FEs	Yes	Yes	Yes	Yes	
Ethnicity FEs	Yes	Yes	Yes	Yes	
Gender FEs	Yes	Yes	Yes	Yes	
<i>B. Subsets of Drugs</i>					
	Tested in Clinical Trials, 2000-2005		Not Tested in Clinical Trials, 1990-2005		
	# Prescriptions in Young Diseases (5)	# Prescriptions in Old Diseases (6)	# Prescriptions in Young Diseases (7)	# Prescriptions in Old Diseases (8)	
Post $\times \mathbf{1}\{Elderly\}$	0.0392* (0.0183)	0.172*** (0.0221)	-0.00749 (0.0124)	-0.00296 (0.0250)	
Mean of Dep. Var.	0.498	0.634	0.225	0.646	
Diff. Wald Test p -value	0.00		0.86		
Observations	53,539	53,539	53,539	53,539	
Adjusted R^2	0.069	0.057	0.011	0.038	
Elderly FE	Yes	Yes	Yes	Yes	
Race FEs	Yes	Yes	Yes	Yes	
Ethnicity FEs	Yes	Yes	Yes	Yes	
Gender FEs	Yes	Yes	Yes	Yes	

NOTE: This table presents coefficients from a patient level regression regresses the outcome in each column on an indicator for after the Medicare Memorandum of 2000 (“Post”) interacted with an indicator for whether the respondents are over 65 (“ $\mathbf{1}\{Elderly\}$ ”). We control for an indicator for elderly, year fixed effects (which encompass the “Post” indicator) and person-level controls for race, ethnicity, and gender. In Panel A, the outcome (# of Prescriptions) uses all drugs. In Panel B, the outcome uses subsets of drugs depending on whether they were tested in clinical trials during the memorandum. Coefficients estimated using seemingly unrelated regressions, which permits a statistical comparison of the coefficients. High Income refers to individuals above 400 percent of the federal poverty line. Standard errors are clustered at the ICD-9 code level and reported in parentheses, with * $p < 0.05$; ** $p < 0.01$ and *** $p < 0.001$.

Table 7: IMPACT ON TRIAL OUTCOMES AND ADVERSE EVENTS

<i>Panel A. Disease-Year Level Analysis, Impact on # Trials</i>				
	Trials With P-value < 0.05		Trials With P-value $\geq$ 0.05	
	(1)		(2)	
Post $\times$ $\mathbf{1}\{MedicareShare\}$	0.0418 (0.0230)		0.0249** (0.00951)	
Mean of Dep. Var.	0.063		0.022	
Diff. Wald Test p -value		0.44		
Observations	4,290		4,290	
Adjusted R^2	0.17		0.134	
ICD9 FEs	Yes		Yes	
ICD9 Category $\times$ Year Trends	Yes		Yes	

<i>Panel B. Drug-Age Category-Gender-Year Level Analysis, Impact on Adverse Event Rate</i>				
	Adverse Event Count		Adverse Event Rate	
	Old Diseases (3)	Young Diseases (4)	Old Diseases (5)	Young Diseases (6)
Post $\times$ $\mathbf{1}\{Elderly\}$	-26.04 (29.63)	48.93 (69.17)	-0.125 (0.0683)	0.0175 (0.0478)
Mean of Dep. Var.	69.109	45.114	1.506	1.236
Diff. Wald Test p -value		0.32		0.09
Observations	21,202	21,202	21,202	21,202
Adjusted R^2	0.296	0.536	0.129	0.033
Year FEs	Yes	Yes	Yes	Yes
Gender FEs	Yes	Yes	Yes	Yes
ICD9 FEs	Yes	Yes	Yes	Yes

NOTE: Panel A presents coefficients from regressing the outcome in each column on an indicator for after the Medicare Memorandum of 2000 (“Post”) interacted with an indicator for whether a disease has an above median share of diagnoses among the elderly (“ $\mathbf{1}\{MedicareShare\}$ ”). Observations are at the disease-year level and we control for ICD-9 code fixed effects (which encompass “ $\mathbf{1}\{Elderly\}$ ”), year fixed effects (which encompass the “Post” indicator) and the ICD-9 category by year trends. Panel B presents coefficients from regressing the outcome in each column on an indicator for “Post” interacted with an indicator for whether the respondents are over 65 (“ $\mathbf{1}\{Elderly\}$ ”). Observations are at the drug by age category by sex by year level. The outcome is the adverse event count in old diseases (Column 3) or young diseases (Column 4). Alternately the outcome is the number of adverse events reported per 1,000 prescriptions in old diseases (Column 5) or young diseases (Column 6). We control for an indicator for elderly, gender, and ICD-9 code. Results are estimated using seemingly unrelated regressions. Standard errors are clustered at the ICD-9 code level and reported in parentheses, with * $p < 0.05$; ** $p < 0.01$ and *** $p < 0.001$.

A Supplemental Appendix

A.1 MEPS Utilization Construction

In our analysis of drug utilization, we link MEPS prescriptions to Cortellis drug names, which provide information on associated diseases and the number of clinical trials over time. We begin with the Cortellis data, which list alternate names for each drug. We group all names ever used for the same molecule together. For example, zoloft, sertraline, sertraline hydrochloride, sertraline hcl, lustral, gladem, tatig, aremis, and CP-51974 are all brand, generic, or development names for sertraline.

We assign each drug group to the disease for which the FDA initially approved it. If there are multiple initial approvals, we use the most common disease listed in Cortellis. For instance, all sertraline-related names are assigned to ICD-9 code 296 (episodic mood disorder).

We also compute the number of clinical trials for each drug group. We restrict to interventional, industry-sponsored trials with at least one U.S. site, conducted between 1990 and 2005. We include all phases, as Phase IV trials may still affect prescription demand. For each drug group, we calculate the total number of trials during the full sample period and the number of trials conducted after the June 2000 memorandum.

Finally, we fuzzy match MEPS drug names to the full list of Cortellis names, keeping matches with a score of at least 0.95 (on a 0 to 1 scale). When multiple matches are tied, we select the shortest Cortellis name. We use this match to assign a disease and number of trials to each MEPS prescription.

A.2 Adverse Events Rate Construction

We examine adverse events using data from the Adverse Event Reporting System (AERS). This section describes the construction of the adverse event rate.

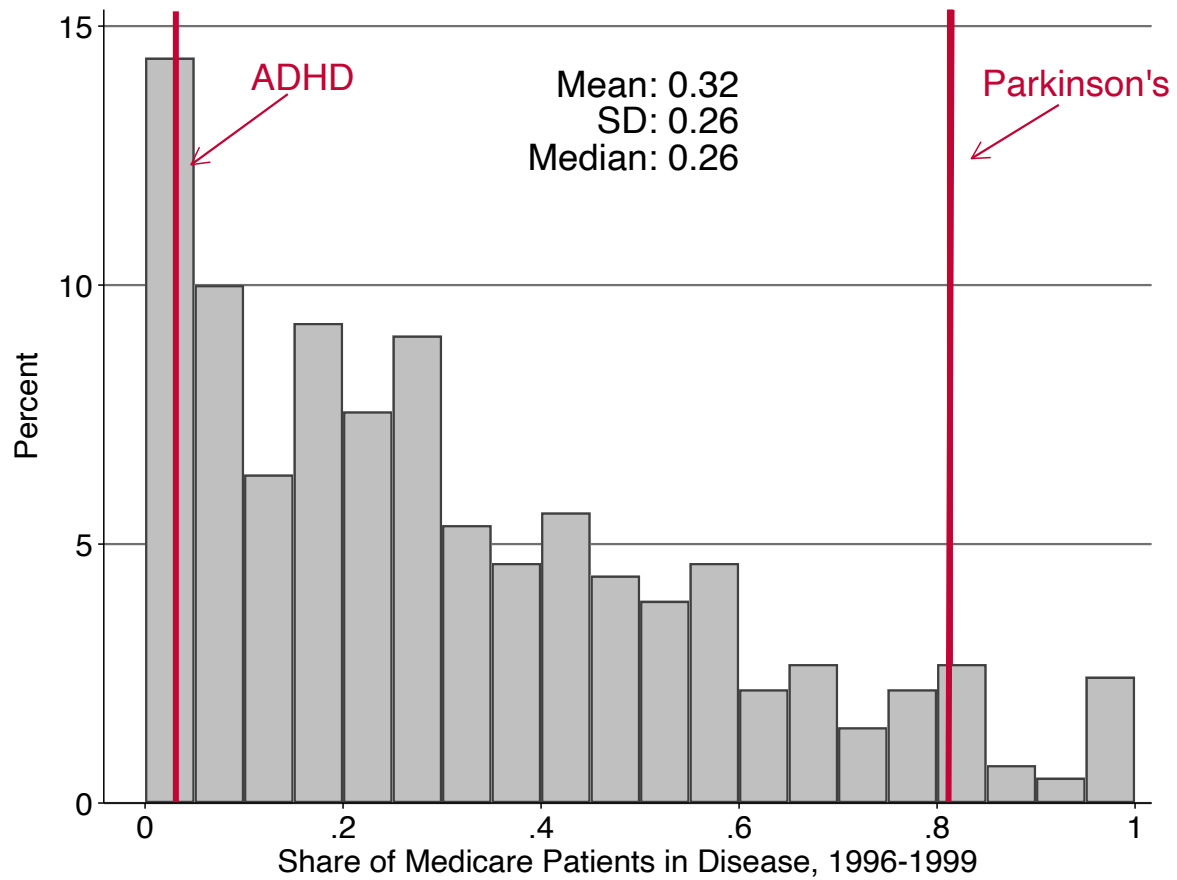
We first construct the denominator of total prescriptions using the MEPS. For each drug, year, age category, and gender, we count the number of unique individuals taking the prescription and use MEPS weights to scale the counts to the U.S. population. We define two age categories: elderly

(ages 65–75) and a control group (ages 50–60).

We then count the number of adverse events reported in AERS for each drug–age–year–gender cell. We merge these counts to MEPS using a fuzzy match on drug name and exact matches on year, age category, and gender. We keep matches with a score above 0.95 (on a 0 to 1 scale). MEPS prescriptions with no matched adverse events are assigned an adverse event rate of zero for that cell. Our main outcome is the number of adverse events per 1,000 unique prescriptions. As in Section 5.2, we compute separate rates for prescriptions matched to either old and young diseases.

A.3 Additional Tables and Figures

Figure A.1: DISEASE-LEVEL EXPOSURE TO MEDICARE POLICY



NOTE: Figure plots the share of diagnoses in a disease among Medicare patients. The sample used is the MEPS from 1996-1999.

Figure A.2: CLINICAL TRIAL ENROLLMENT DATA FROM CLINICALTRIALS.GOV

COMPLETED ⓘ

Efficacy and Safety of Adalimumab in Patients With Active Rheumatoid Arthritis Treated Concomitantly With Methotrexate.

ClinicalTrials.gov ID ⓘ NCT00195702

Sponsor ⓘ Abbott

Information provided by ⓘ Abbott (Responsible Party)

AGE, CUSTOMIZED

Measure Type: Number | Unit of measure: participants

Number Analyzed	619 participants
< 40 years	55
Between 40 and 64 years	402
Between 65 and 74 years	130
>= 75 years	32

STUDY START ⓘ

2000-02

PRIMARY COMPLETION (ACTUAL) ⓘ

2002-09

STUDY COMPLETION (ACTUAL) ⓘ

2010-08

ENROLLMENT (ACTUAL) ⓘ

619

STUDY TYPE ⓘ

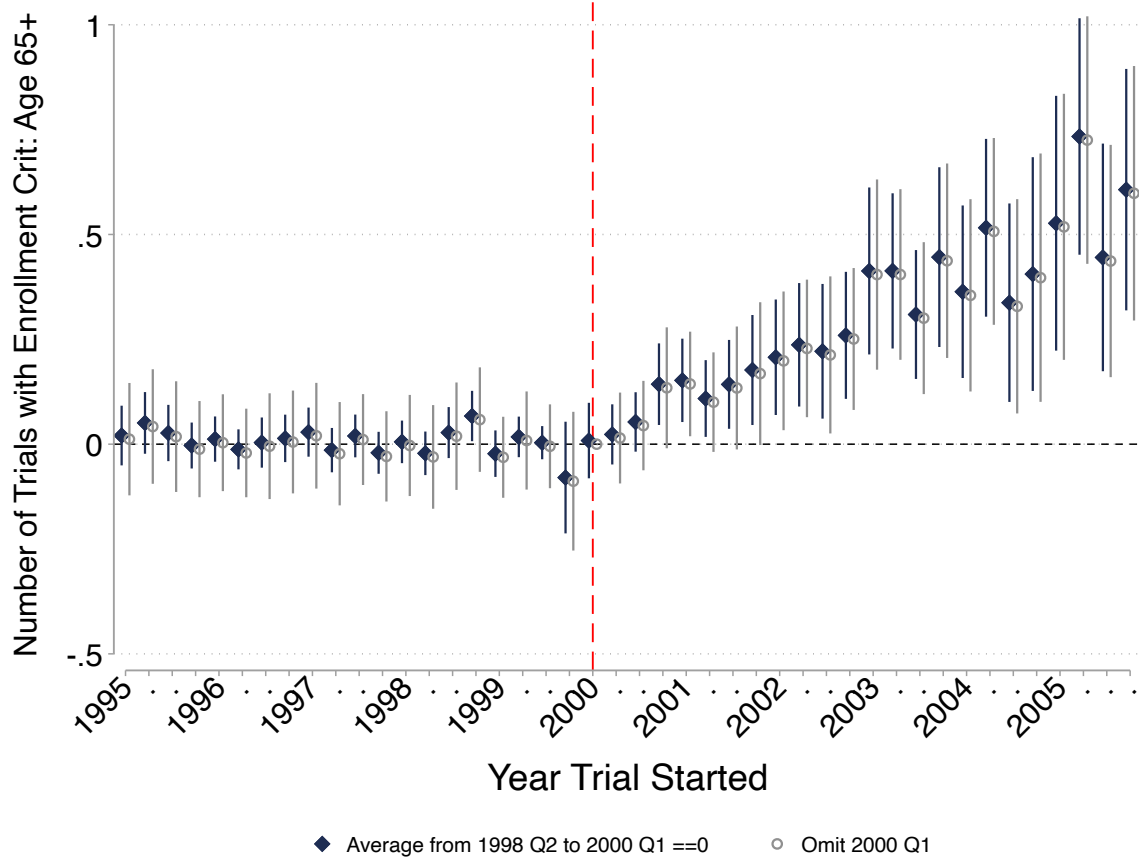
Interventional

PHASE ⓘ

Phase 3

NOTE: This figure shows an example of trial enrollment counts in ClinicalTrials.gov. This trial was a phase 3 trial testing the efficacy of adalimumab. The trial was conducted in 2000, and adalimumab (brand name Humira) was approved by the FDA in 2002.

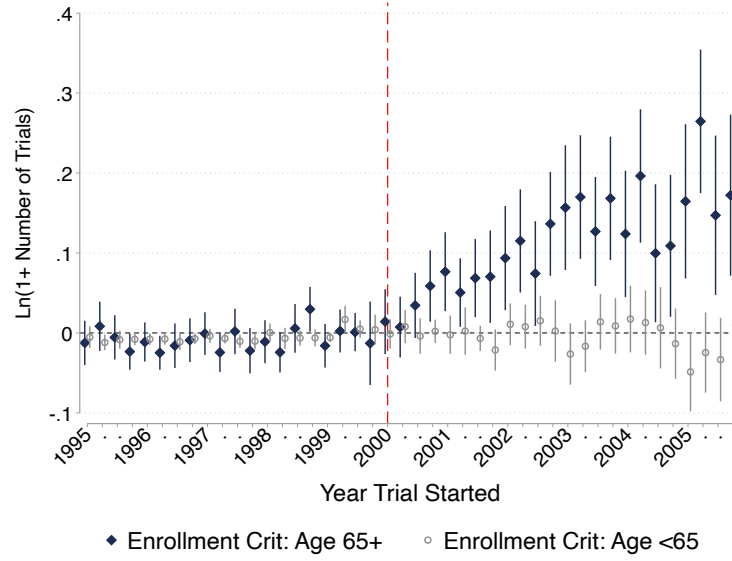
Figure A.3: EFFECT ON TRIALS WITH ELDERLY IN ENROLLMENT CRITERIA,
CHANGE IN OMITTED GROUP



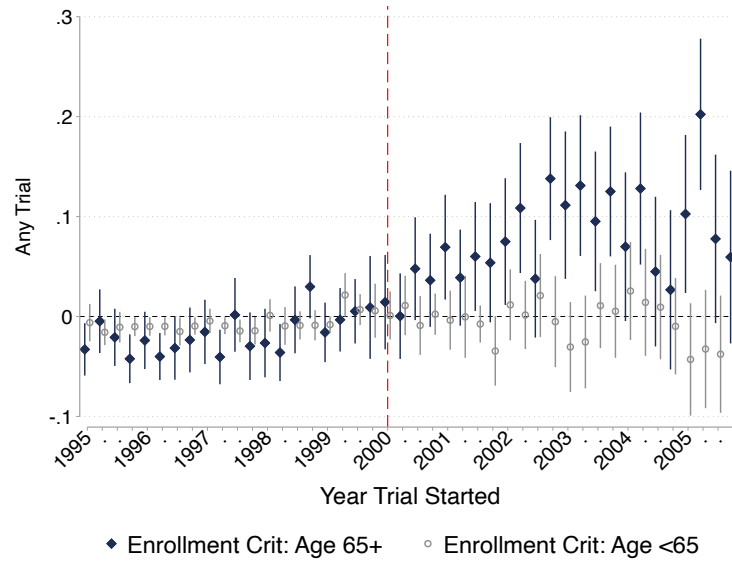
NOTE: This figure is similar to Panel A of Figure 2, but changes the omitted group in the equation estimation. For ease of interpretation, only the number of trials that include the elderly participants in their enrollment criteria are included in the figure. Observations are at the disease-quarter level and the outcome is the number of trials that include the elderly in their enrollment criteria in a disease-quarter. The estimates in black diamonds are the same as in the specification used in the main body of the paper and the estimates in hollow circles omits just the quarter before the policy. Solid vertical lines are 95 percent confidence intervals. The dotted vertical line represents the quarter before the memorandum.

**Figure A.4: EFFECT ON ENROLLMENT CRITERIA,
WITH LOG TRIALS AND ANY TRIAL**

(a) Log Number of Trials

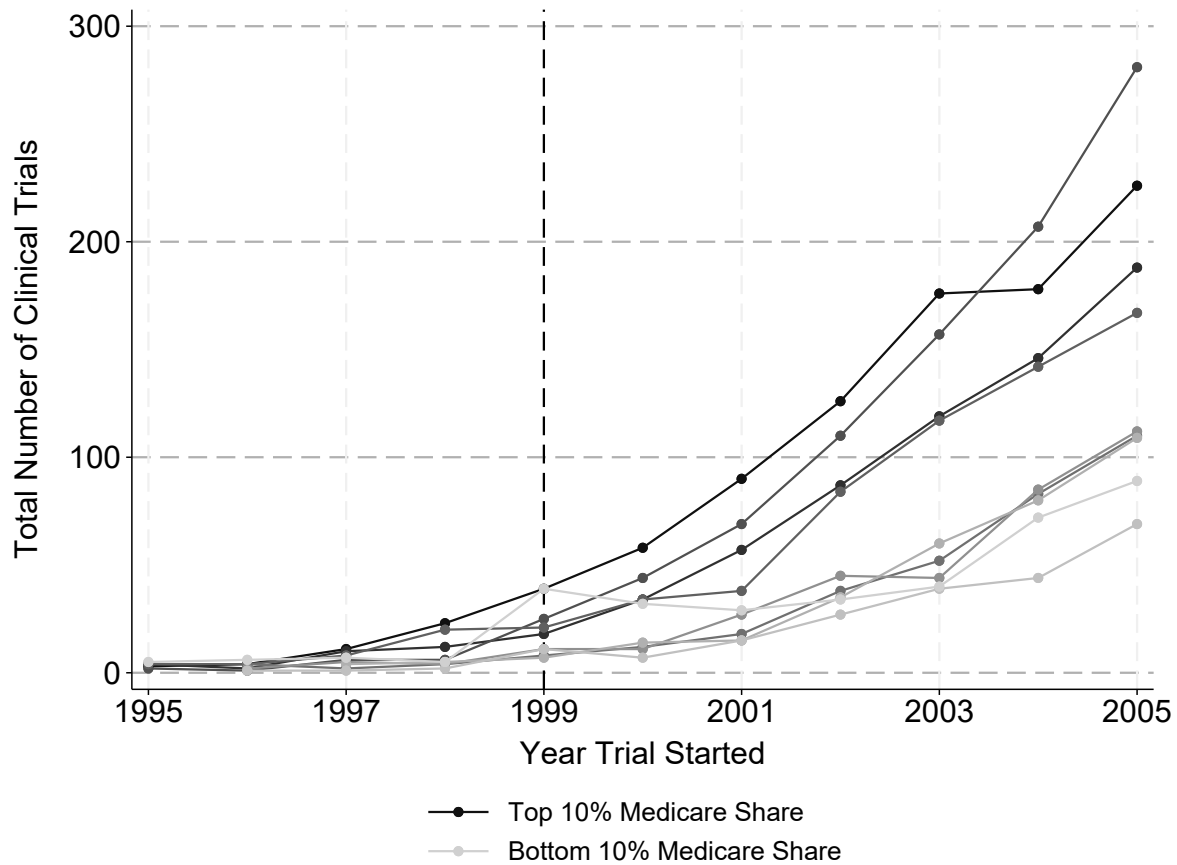


(b) Any Trial



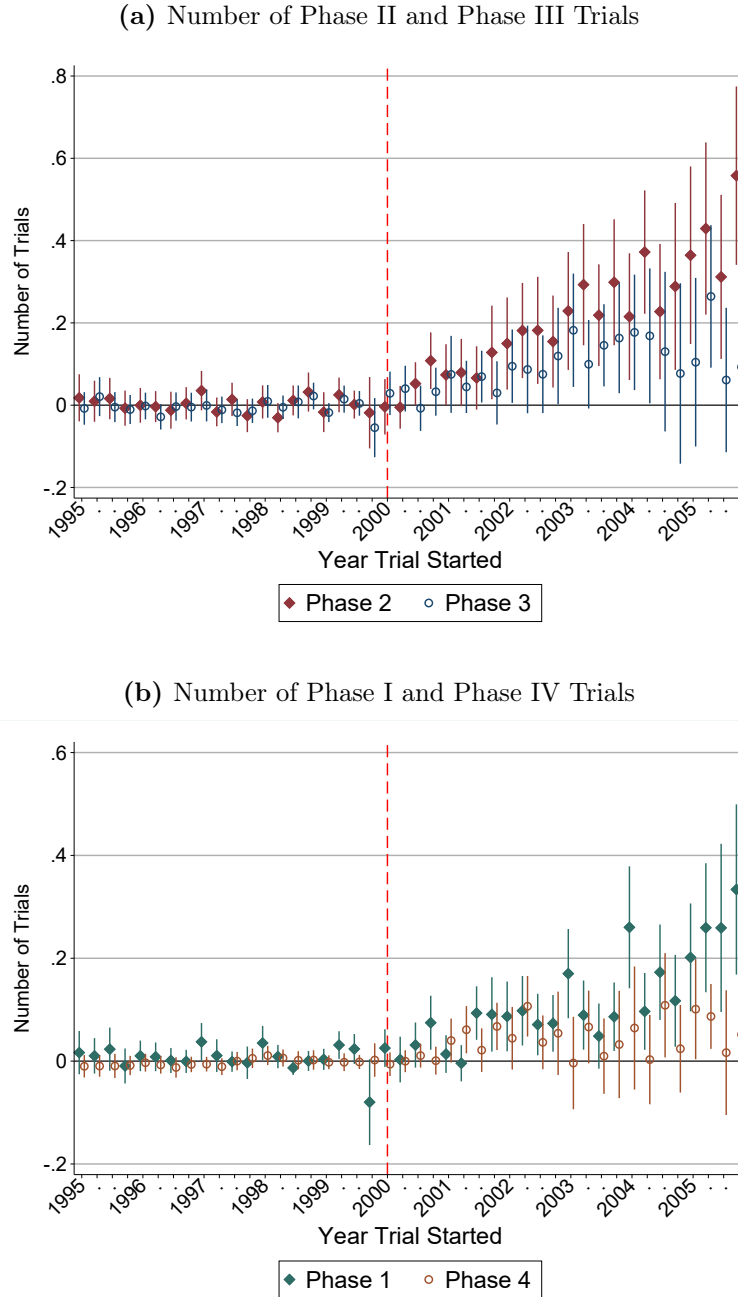
NOTE: This figure is similar to Panel A of Figure 2, but examines the impact on the log number of clinical trials, by enrollment criteria (Panel A) and the likelihood of any clinical trial, by enrollment criteria (Panel B). Observations are at the disease-quarter level. We plot the coefficients on the interaction between quarter and an indicator for whether a disease had an above median share of diagnoses among the elderly from 1996-1999. Solid vertical lines are 95 percent confidence intervals. The dotted vertical line represents the quarter before the memorandum.

Figure A.5: DESCRIPTIVE EVIDENCE OF MEDICARE POLICY



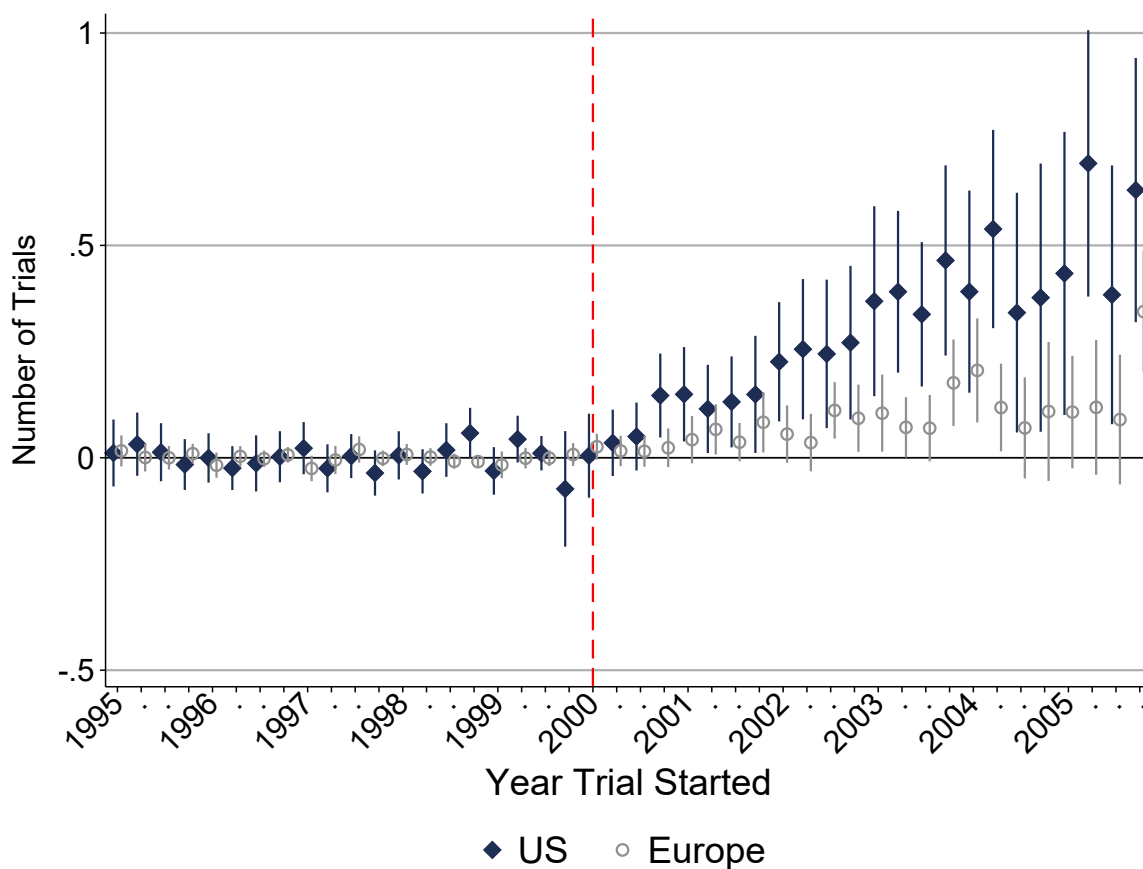
NOTE: This figure plots the total number of clinical trials within disease categories. The darkest line refers to diseases in the top 10 percent of Medicare share of diagnoses. The second-darkest contains the diseases in the top 10-20 percent of Medicare share of diagnoses all the way down to the lightest line which contains the lowest 10 percent of diseases in terms of the Medicare share of diagnoses. The increase in the number of trials for diseases in the bottom 10 percent Medicare share in 1999 is primarily driven by HIV-related clinical trials. The dotted vertical line represents the year before the memorandum.

Figure A.6: IMPACT ON NUMBER OF TRIALS, BY TRIAL PHASE



NOTE: This figure is similar to Figure 3, but examines differences across trial phases. Observations are at the disease-quarter level. We plot the coefficients on the interaction between quarter and an indicator for whether a disease had an above median share of diagnoses among the elderly from 1996-1999. The outcomes are the number of Phase I, Phase II, Phase III, and Phase IV trials. Solid vertical lines are 95 percent confidence intervals. The dotted vertical line represents the quarter before the memorandum.

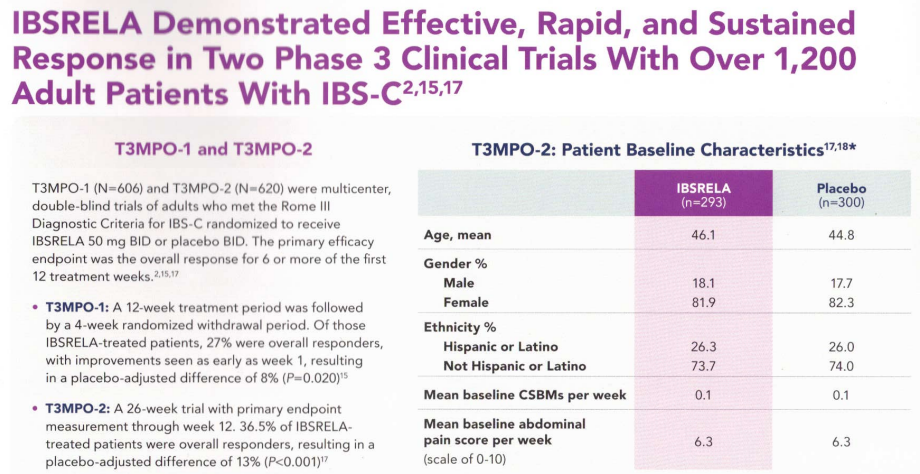
Figure A.7: IMPACT ON NUMBER OF TRIALS IN U.S. vs. EUROPE



NOTE: This figure presents event study coefficients from the estimation of equation (1) separately for trials in the United States versus Europe. Observations are at the disease-quarter level and the outcome is the number of trials in a disease-quarter. We plot the coefficients on the interaction between year and an indicator for whether a disease had an above median share of diagnoses among the elderly from 1996-1999. Solid vertical lines are 95 percent confidence intervals. The dotted vertical line represents the quarter before the memorandum.

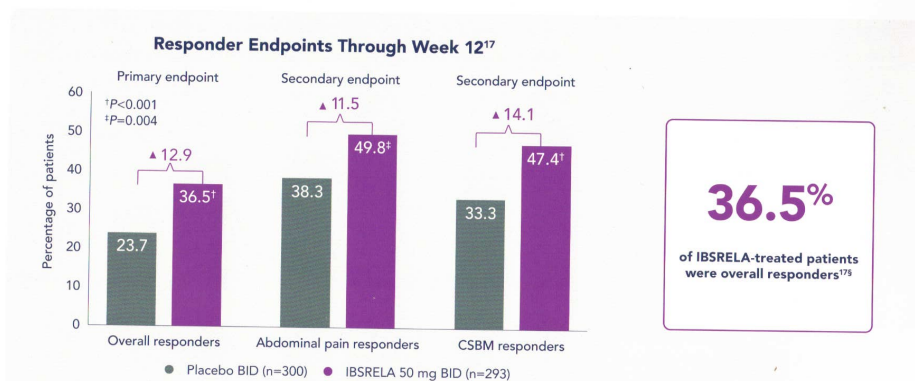
Figure A.8: EXAMPLE MARKETING MATERIALS

(a)



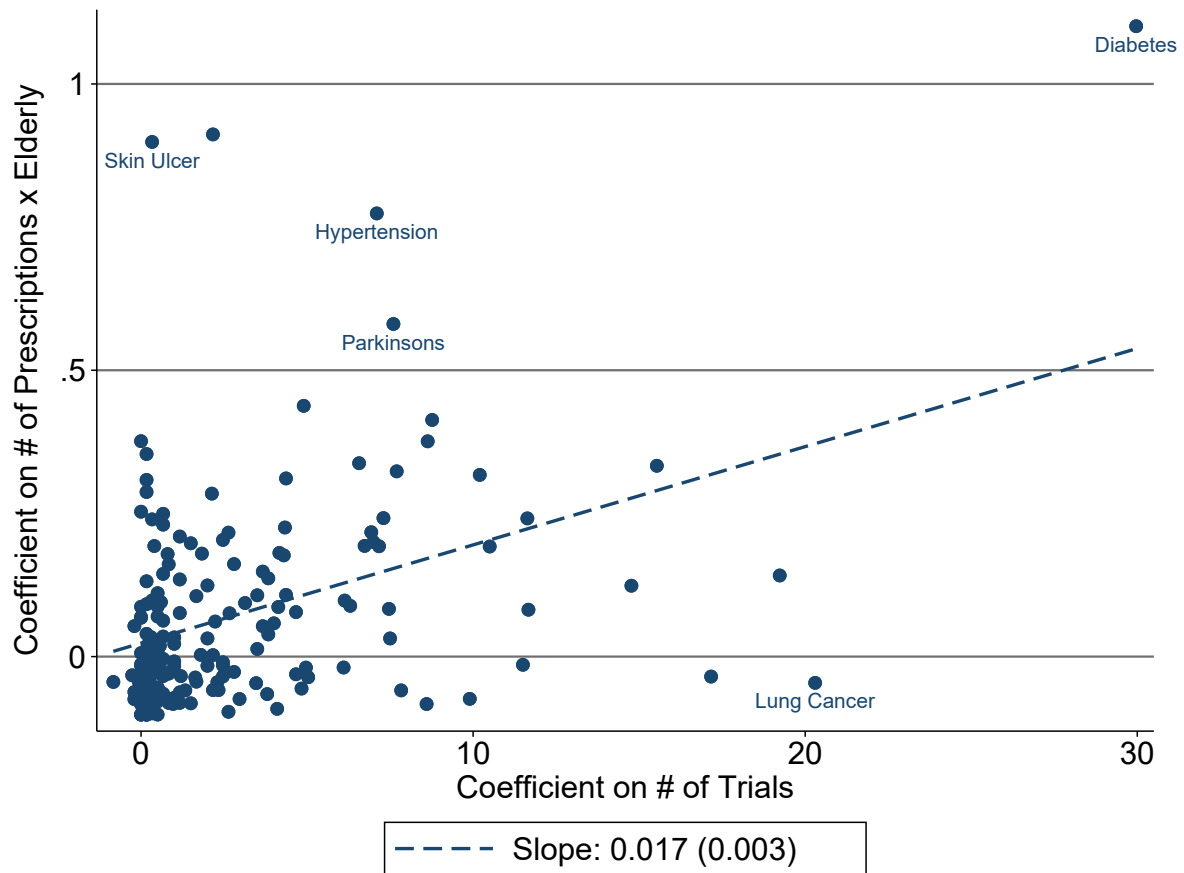
(b)

Significantly More IBSRELA-Treated Patients Were Overall Responders in T3MPO-2¹⁷



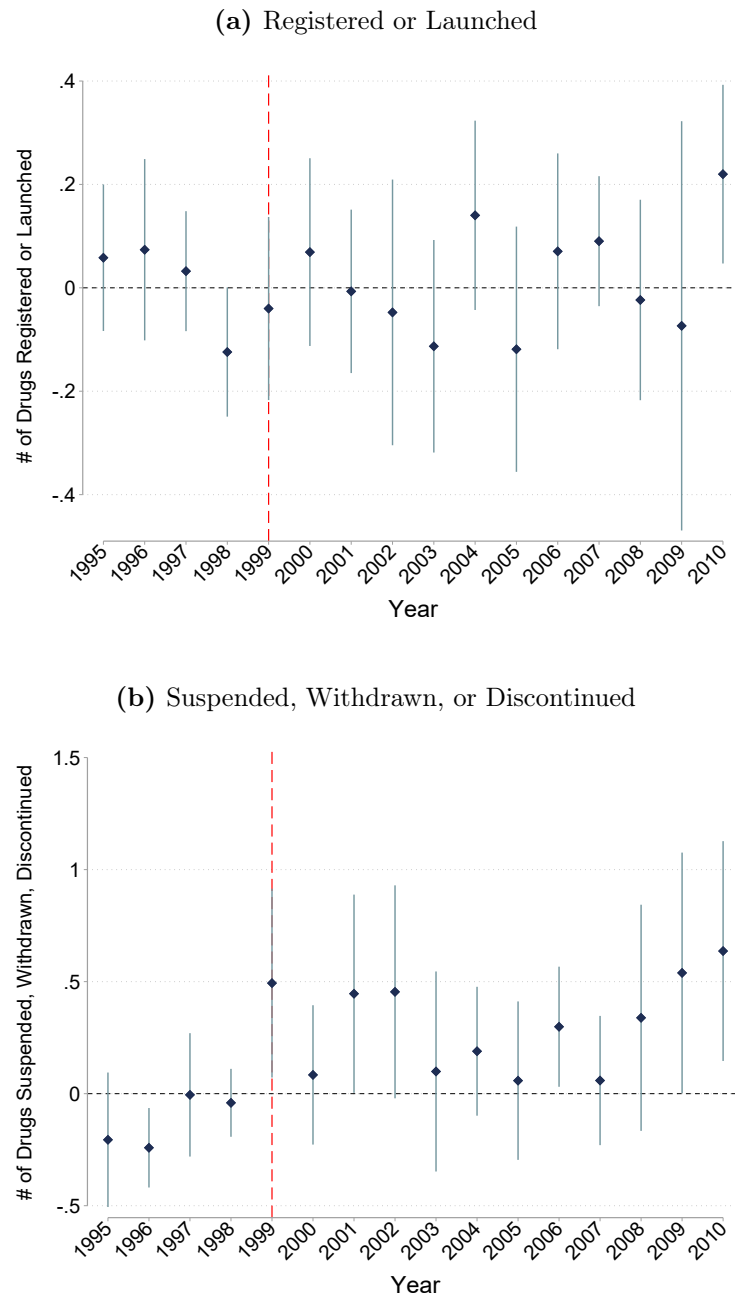
NOTE: This figure shows two pages of excerpts from marketing materials for IBSRELA (tenapanor) which is approved for the treatment of irritable bowel syndrome. Print copies were delivered to a gastroenterologist office in Ohio.

Figure A.9: DISEASE-LEVEL CORRELATION BETWEEN TRIALS AND PRESCRIPTIONS



NOTE: This figure presents the correlation between the change in the number of trials and the change in the number of prescriptions at the ICD-9 code level. We regress the number of trials in each disease and year on an indicator for after the Medicare memorandum, an indicator for each disease and the interaction. The x-axis plots the coefficient on the interaction for each ICD-9 code. This represents how much the number of trials has increases in this disease after the memorandum. In our second regression the outcome is the number of prescriptions in a person by disease by year. We regress this on an indicator for after the Medicare memorandum, an indicator for each disease, and indicator for whether the individual is over 65 and all the interactions thereof and control for race, Hispanic, and sex. We use analytical MEPS weights. The y-axis plots the coefficient on the interaction for after the memorandum, whether the individual is over 65 and each ICD-9 code which represents how much the number of prescriptions increases for elderly individuals in this disease after the memorandum.

Figure A.10: DRUG APPROVALS AND TERMINATIONS



NOTE: This figure presents event study coefficients from the estimation of equation (1). Observations are at the disease-quarter level. We plot the coefficients on the interaction between quarter and an indicator for whether a disease had an above median share of diagnoses among the elderly from 1996-1999. The outcome in Panel A is the number of new drugs that were approved (i.e., registered or launched) that quarter. The outcome in Panel B is the number of drugs that were terminated (i.e., suspended, withdrawn, or discontinued) in that quarter. Solid vertical lines are 95 percent confidence intervals. The dotted vertical line represents the quarter before the memorandum.

Table A.1: EFFECT ON ENROLLMENT (1995 TO 2003)

	# Trials Enrolling Only Age < 65) (1)	# Trials Enrolling Any Age ≥ 65) (2)	Share of Participants Age ≥ 65 (3)
Post × $\mathbf{1}\{MedicareShare\}$	0.0113 (0.0360)	0.858*** (0.172)	0.305 (0.179)
Mean of Dep. Var.	0.081	0.556	0.112
Diff. Wald Test p -value		0.00	
Observations	3,510	3,510	170
Adjusted R^2	0.208	0.479	0.232
Year FEs	Yes	Yes	Yes
ICD9 FEs	Yes	Yes	Yes
ICD9 Category × Year Trends	Yes	Yes	Yes

NOTE: This table presents coefficients from regressing the outcome in each column on an indicator for after the Medicare Memorandum of 2000 (“Post”) interacted with an indicator for whether a disease has an above median share of diagnoses among the elderly (“ $\mathbf{1}\{MedicareShare\}$ ”). Observations are at the disease-year level. We control for ICD-9 code fixed effects (which encompass (“ $\mathbf{1}\{MedicareShare\}$ ”)), year fixed effects (which encompass the “Post” indicator) and the ICD-9 category by year trends. Columns 1 and 2 estimated using seemingly unrelated regressions. The “Diff. Wald Test p -value” is the p -value comparing the $Post \times MedicareShare$ coefficient in Columns 1 vs. 2. Column 3 is estimated using OLS. Standard errors are clustered at the ICD-9 code level and reported in parentheses, with * $p < 0.05$; ** $p < 0.01$ and *** $p < 0.001$.

Table A.2: EFFECT ON ENROLLMENT, COMPARISON BETWEEN MEDICARE MEMORANDUM OF 2000 AND MEDICARE PART D

	# Trials Enrolling Only Age < 65) (1)	# Trials Enrolling Any Age ≥ 65) (2)	Share of Participants Age ≥ 65 (3)
Post2000 × $\mathbf{1}\{MedicareShare\}$	0.0350 (0.0410)	0.780*** (0.181)	0.183 (0.106)
PostPartD × $\mathbf{1}\{MedicareShare\}$	-0.0807 (0.0851)	1.123*** (0.295)	-0.00394 (0.0373)
Mean of Dep. Var.	0.148	0.941	0.129
Diff. Wald Test p -value	0.00		
Observations	4,290	4,290	409
Adjusted R^2	0.295	0.512	0.330
Year FEs	Yes	Yes	Yes
ICD9 FEs	Yes	Yes	Yes
ICD9 Category × Year Trends	Yes	Yes	Yes

NOTE: This table presents coefficients from regressing the outcome in each column on an indicator for after the Medicare Memorandum of 2000 (“Post2000”) interacted with an indicator for whether a disease has an above median share of diagnoses among the elderly (“ $\mathbf{1}\{MedicareShare\}$ ”). We also include an indicator for after part D was announced in 2003 (“PostPartD”) interacted with (“ $\mathbf{1}\{MedicareShare\}$ ”). Observations are at the disease-year level. We control for ICD-9 code fixed effects (which encompass “ $\mathbf{1}\{MedicareShare\}$ ”), year fixed effects (which encompass the “Post2000” and “PostPartD” indicators) and the ICD-9 category by year trends. Columns 1 and 2 estimated using seemingly unrelated regressions. The “Diff. Wald Test p -value” is the p -value comparing the $Post \times MedicareShare$ coefficient in Columns 1 vs. 2. Column 3 is estimated using OLS. Standard errors are clustered at the ICD-9 code level and reported in parentheses, with * $p < 0.05$; ** $p < 0.01$ and *** $p < 0.001$.

Table A.3: IMPACT ON ENROLLMENT, EXCLUDING FAST-TRACK DISEASES

	# Trials Enrolling Only Age < 65) (1)	# Trials Enrolling Any Age ≥ 65) (2)	Share of Participants Age ≥ 65 (3)
Post × $\mathbf{1}\{MedicareShare\}$	-0.0343 (0.0554)	0.703** (0.217)	0.115 (0.141)
Mean of Dep. Var.	0.118	0.653	0.123
Diff. Wald Test p -value	0.00		
Observations	3,916	3,916	299
Adjusted R^2	0.279	0.474	0.182
Year FEs	Yes	Yes	Yes
ICD9 FEs	Yes	Yes	Yes
ICD9 Category × Year Trends	Yes	Yes	Yes

NOTE: This table presents coefficients from regressing the outcome in each column on an indicator for after the Medicare Memorandum of 2000 (“Post”) interacted with an indicator for whether a disease has an above median share of diagnoses among the elderly (“ $\mathbf{1}\{MedicareShare\}$ ”). Observations are at the disease-year level. ICD-9 codes associated with drugs receiving a Fast-Track designation are excluded. We control for ICD-9 code fixed effects (which encompass “ $\mathbf{1}\{MedicareShare\}$ ”), year fixed effects (which encompass the “Post” indicator) and ICD-9 category by year trends. Columns 1 and 2 estimated using seemingly unrelated regressions. The “Diff. Wald Test p -value” is the p -value comparing the $Post \times MedicareShare$ coefficient in Columns 1 vs. 2. Column 3 is estimated using OLS. Standard errors are clustered at the ICD-9 code level and reported in parentheses, with * $p < 0.05$; ** $p < 0.01$ and *** $p < 0.001$.

Table A.4: HETEROGENEITY ACROSS DISEASES

	Dependent Variable: # Trials			
	Duration		Patient Care Type	
	Short Trials (1)	Long Trials (2)	Outpatient (3)	Inpatient (4)
$\text{Post} \times \mathbf{1}\{\text{MedicareShare}\}$	0.575* (0.261)	1.770*** (0.484)	0.397 (0.208)	0.896 (0.482)
Mean of Dep. Var.	0.714	1.444	0.463	1.744
Diff. Wald Test p -value	0.03		0.34	
Observations	2,332	1,958	2,255	2,035
Adjusted R^2	0.436	0.552	0.419	0.570
Year FEs	Yes	Yes	Yes	Yes
ICD9 FEs	Yes	Yes	Yes	Yes
ICD9 Category $\times$ Year Trends	Yes	Yes	Yes	Yes

NOTE: This table presents coefficients from regressing the number of trials on an indicator for after the Medicare Memorandum of 2000 (“Post”) interacted with an indicator for whether a disease has an above median share of diagnoses among the elderly (“ $\mathbf{1}\{\text{MedicareShare}\}$ ”). Observations are at the disease-year level. In Column 1 we restrict the sample to diseases with a below median average trial duration from 1995-1999. In Column 2 we restrict the sample to diseases with an above median trial duration. We next divide diseases into those that are below median in terms of representation among MEPS inpatient data from 1996-1999 (Column 3) and above median (Column 4). We control for ICD-9 code fixed effects (which encompass “ $\mathbf{1}\{\text{MedicareShare}\}$ ”), year fixed effects (which encompass the “Post” indicator) and the ICD-9 category by year trends. Results are estimated using seemingly unrelated regressions. Standard errors are clustered at the ICD-9 code level and reported in parentheses, with * $p < 0.05$; ** $p < 0.01$ and *** $p < 0.001$.

Table A.5: EFFECT ON NUMBER OF TRIALS,
ACCOUNTING FOR CONFOUNDING POLICIES

	Dependent Variable: # Trials		
	Restricting to 1995 to 2003 (1)	Including Controls For Post Part D (2)	Excluding Fast Track Diseases (3)
Post $\times$ $\mathbf{1}\{MedicareShare\}$	0.869*** (0.190)	0.815*** (0.201)	0.668** (0.248)
Observations	3,510	4,290	3,916
Mean of Dep. Var.	1.051	1.051	0.747
Adjusted R^2	0.484	0.513	.467
Year FEs	Yes	Yes	Yes
ICD9 FEs	Yes	Yes	Yes
ICD9 $\times$ Year Trends	Yes	Yes	Yes

NOTE: This table presents coefficients from regressing the number of trials on an indicator for after the Medicare Memorandum of 2000 (“Post”) interacted with an indicator for whether a disease has an above median share of diagnoses among the elderly (“ $\mathbf{1}\{MedicareShare\}$ ”). Observations are at the disease-year level. Column 1 restricts the same to 1995 to 2003. Column 2 includes an indicator for after Part D was announced in 2003 and the interaction of this with (“ $\mathbf{1}\{MedicareShare\}$ ”). Column 3 excludes ICD-9 codes associated with drugs receiving a Fast-Track designation. We control for ICD-9 code fixed effects (which encompass “ $\mathbf{1}\{MedicareShare\}$ ”), year fixed effects (which encompass the “Post” indicator) and the ICD-9 category by year trends. Estimates are from OLS regressions. Standard errors are clustered at the ICD-9 code level and reported in parentheses, with * $p < 0.05$; ** $p < 0.01$ and *** $p < 0.001$.